1. Determine whether the statement is true or false :

(A) $\lim\limits_{x\to4}\left(\frac{2x}{x-4}-\frac{8}{x-4}\right)=\lim\limits_{x\to4}\frac{2x}{x-4}-\lim\limits_{x\to4}\frac{8}{x-4}$ (B) $\lim\limits_{x\to1}\frac{x^2+6x-7}{x^2+5x-6}=\frac{\lim\limits_{x\to1}(x^2+6x-7)}{\lim\limits_{x\to1}(x^2+5x-6)}$

(C) $\lim\limits_{x\to1}\frac{x-3}{x^2+2x-4}=\frac{\lim\limits_{x\to1}(x-3)}{\lim\limits_{x\to1}(x^2+2x-4)}$

(D) If $\lim\limits_{x\to5}f(x)=2$ and $\lim\limits_{x\to5}g(x)=0$, then $\lim\limits_{x\to5}\frac{f(x)}{g(x)}$ does not exist.

(E) If $\lim\limits_{x\to5}f(x)=0$ and $\lim\limits_{x\to5}g(x)=2$, then $\lim\limits_{x\to5}\frac{f(x)}{g(x)}$ does not exist.

सत्य/असत्य :

(A) $\lim\limits_{x\to4}\left(\frac{2x}{x-4}-\frac{8}{x-4}\right)=\lim\limits_{x\to4}\frac{2x}{x-4}-\lim\limits_{x\to4}\frac{8}{x-4}$

(B) $\lim\limits_{x\to1}\frac{x^2+6x-7}{x^2+5x-6}=\frac{\lim\limits_{x\to1}(x^2+6x-7)}{\lim\limits_{x\to1}(x^2+5x-6)}$

(C) $\lim\limits_{x\to1}\frac{x-3}{x^2+2x-4}=\frac{\lim\limits_{x\to1}(x-3)}{\lim\limits_{x\to1}(x^2+2x-4)}$

(D) यदि $\lim\limits_{x\to5}f(x)=2$ तथा $\lim\limits_{x\to5}g(x)=0$, तो $\lim\limits_{x\to5}\frac{f(x)}{g(x)}$ विद्यमान नहीं होगी

(E) यदि $\lim\limits_{x\to5}f(x)=0$ तथा $\lim\limits_{x\to5}g(x)=2$, तो $\lim\limits_{x\to5}\frac{f(x)}{g(x)}$ विद्यमान नहीं होगी

**Ans.** (a) False (b) False (c) True (d) True (e) False

**Sol.** (A) False

$\lim\limits_{x\to4}\left(\frac{2x}{x-4}-\frac{8}{x-4}\right)=2$ but $\lim\limits_{x\to4}\frac{2x}{x-4}=\text{DNE}$

(B) False

$$\lim_{x\to 1}\frac{x^2+6x-7}{x^2+5x-6}=\lim_{x\to 1}\frac{(x-1)(x+7)}{(x-1)(x+6)}=\frac{8}{7}$$

but $\dfrac{\lim\limits_{x\to 1}x^2+6x-7}{\lim\limits_{x\to 1}x^2+5x-6}=\text{DNE}$

(C) $\lim\limits_{x\to 1}\dfrac{x-3}{x^2+2x-4}=\dfrac{\lim\limits_{x\to 1}x-3}{\lim\limits_{x\to 1}x^2+2x-4}=\dfrac{-2}{-1}=2$ (True)

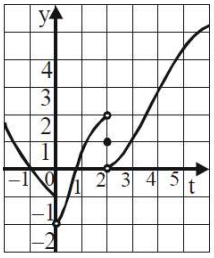
(D) $\lim\limits_{x\to 5}f(x)=2,\ \lim\limits_{x\to 5}g(x)=0$

$\Rightarrow \lim\limits_{x\to 5}\dfrac{f(x)}{g(x)}=\text{DNE}$ (True)

(E) $\lim\limits_{x\to 5}f(x)=0$ & $\lim\limits_{x\to 5}g(x)=2$

$\Rightarrow \lim\limits_{x\to 5}\dfrac{f(x)}{g(x)}=0$ (False)

2. For the function g whose graph is given, state the value of each quantity, if it exists :



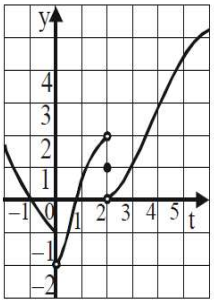

(A) $\lim\limits_{t\to 0^-}g(t)$ (B) $\lim\limits_{t\to 0^+}g(t)$

(C) $\lim\limits_{t\to 0}g(t)$ (D) $\lim\limits_{t\to 2^-}g(t)$

(E) $\lim\limits_{t\to 2^+}g(t)$ (F) $\lim\limits_{t\to 2}g(t)$

(G) $g(2)$ (H) $\lim\limits_{t\to 4}g(t)$

फलन g का आरेख दिया गया है। निम्न राशियां ज्ञात कीजिए–




(A) $\lim\limits_{t\to 0^-}g(t)$ (B) $\lim\limits_{t\to 0^+}g(t)$

(C) $\lim\limits_{t\to 0}g(t)$ (D) $\lim\limits_{t\to 2^-}g(t)$

(E) $\lim\limits_{t\to 2^+}g(t)$ (F) $\lim\limits_{t\to 2}g(t)$

(G) $g(2)$ (H) $\lim\limits_{t\to 4}g(t)$

**Ans.** (a) –1 (b) –2 (c) does not exist (d) 2 (e) 0
(f) DNE (g) 1 (h) 3

**Sol.** (A) $\lim_{t\to 0^-} g(t) = -1$ (B) $\lim_{t\to 0^+} g(t) = -2$ (C) $\lim_{t\to 0} g(t) = \text{DNE}$ (D) $\lim_{t\to 2^-} g(t) = 2$

(E) $\lim_{t\to 2^+} g(t) = 0$ (F) $\lim_{t\to 2} g(t) = \text{DNE}$ (G) $g(2) = 1$ (H) $\lim_{t\to 4} g(t) = 3$

3. Calculate the following limits :

(i) $\lim_{x\to 1}\left(\frac{1}{1-x} - \frac{3}{1-x^3}\right)$

(ii) $\lim_{x\to 0}\frac{\sqrt{1+x} - \sqrt{1-x}}{2x}$

(iii) $\lim_{x\to 0}\frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$

(iv) $\lim_{n\to\infty}\frac{(n+1)^4 - (n-1)^4}{(n+1)^4 + (n-1)^4}$

(v) $\lim_{n\to\infty}\frac{(n+2)! + (n+1)!}{(n+2)! - (n+1)!}$

(vi) $\lim_{x\to a}\left(\frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}\right)$

(vii) $\lim_{n\to\infty}\left(\frac{1}{1.3} + \frac{1}{3.5} + .... + \frac{1}{(2n-1)(2n+1)}\right)$

(viii) $\lim_{x\to\infty}\left[\frac{3x^2}{2x+1} - \frac{(2x-1)(3x^2+x+2)}{4x^2}\right]$

(ix) $\lim_{n\to\infty}\frac{1}{n^2}(1+2+3+.....+n)$

निम्न की सीमा ज्ञात कीजिए :

(i) $\lim_{x\to 1}\left(\frac{1}{1-x} - \frac{3}{1-x^3}\right)$

(ii) $\lim_{x\to 0}\frac{\sqrt{1+x} - \sqrt{1-x}}{2x}$

(iii) $\lim_{x\to 0}\frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$

(iv) $\lim_{n\to\infty}\frac{(n+1)^4 - (n-1)^4}{(n+1)^4 + (n-1)^4}$

(v) $\lim_{n\to\infty}\frac{(n+2)! + (n+1)!}{(n+2)! - (n+1)!}$

(vi) $\lim_{x\to a}\left(\frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}\right)$

(vii) $\lim_{n\to\infty}\left(\frac{1}{1.3} + \frac{1}{3.5} + .... + \frac{1}{(2n-1)(2n+1)}\right)$

(viii) $\lim_{x\to\infty}\left[\frac{3x^2}{2x+1} - \frac{(2x-1)(3x^2+x+2)}{4x^2}\right]$

(ix) $\lim_{n\to\infty}\frac{1}{n^2}(1+2+3+.....+n)$

**Ans.** (i) –1 (ii) 1/2 (iii) 1/2 (iv) 0

(v) 1 (vi) $\frac{2}{\left(3\sqrt{3}\right)}$ (vii) 1/2 (viii) –1/2 (ix) 1/2

**Sol.** (i) $\lim\limits_{x\to 1}\left(\frac{1}{1-x}\frac{-3}{1-x^3}\right)=\lim\limits_{x\to 1}\frac{x^2+x-2}{(1-x)\left(1+x+x^2\right)}$

$$=\lim_{x\to 1}\frac{(x-1)(x+2)}{(1-x)\left(1+x+x^2\right)}=-1$$

(ii) $\lim\limits_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{2x}=\lim\limits_{x\to 0}\frac{2x}{2x\left(\sqrt{1+x}+\sqrt{1-x}\right)}$

$$=\frac{1}{2}$$

(iii) $\lim\limits_{x\to 0}\frac{\left(1+x^2\right)^{\frac{1}{3}}-(1-2x)^{\frac{1}{4}}}{x(1+x)}\left(\frac{0}{0}\right)$

$$=\lim_{x\to 0}\frac{\left(1+\frac{1}{3}x^2+\frac{1}{3}\left(\frac{1}{3}-1\right)\frac{x^4}{\underline{|2}}+.....\right)-\left(1-\frac{2x}{4}+\frac{1}{4}\left(\frac{1}{4}-1\right)\frac{(2x)^2}{\underline{|2}}+.....\right)}{x(1+x)}$$

$$=\lim_{x\to 0}\frac{\frac{1}{2}+\frac{1}{3}x+.....}{1+x}=\frac{1}{2}$$

(iv) $\lim\limits_{n\to\infty}\frac{\left(1+\frac{1}{n}\right)^4-\left(1-\frac{1}{n}\right)^4}{\left(1+\frac{1}{n}\right)^4+\left(1+\frac{1}{n}\right)^4}=0$

(v) $\lim\limits_{n\to\infty}\frac{\underline{|n+2}+\underline{|n+1}}{\underline{|n+2}-\underline{|n+1}}=\lim\limits_{n\to\infty}\frac{n+3}{n+1}=1$

(vi) $\lim_{x\to a}\left(\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}\right)=\lim_{x\to a}\frac{\sqrt{3a+x}+2\sqrt{x}}{(3a-3x)\left(\sqrt{a+2x}+\sqrt{3x}\right)}$

$=\frac{1}{3}\cdot\frac{4\sqrt{a}}{2\sqrt{3a}}=\frac{2}{3\sqrt{3}}$

(vii) $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{(2r-1)(2r+1)}$

$=\lim_{n\to\infty}\frac{1}{2}\left[\sum_{r=1}^{n}\frac{(2r+1)-(2r-1)}{(2r-1)(2r+1)}\right]$

$=\frac{1}{2}\lim_{n\to\infty}\left[\sum_{r=1}^{n}\left(\frac{1}{2r-1}-\frac{1}{2r+1}\right)\right]$

$=\frac{1}{2}\lim_{n\to\infty}\left[1-\frac{1}{3}+\frac{1}{3}-\frac{1}{5}+.....+\frac{1}{2n-1}-\frac{1}{2+1}\right]$

$=\frac{1}{2}$

(viii) $\lim_{x\to\infty}\left[\frac{3x^2}{2x+1}-\frac{(2x-1)\left(3x^2+x+2\right)}{4x^2}\right]$

$=\lim_{x\to\infty}\frac{12x^4-\left(4x^2-1\right)\left(3x^2+x+2\right)}{\left(2+\frac{1}{x}\right)4x^3}=\lim_{x\to\infty}\frac{\left(-4-\frac{8}{x}+\frac{3}{x}-\frac{1}{x^2}-\frac{2}{x}\right)}{4\left(2+\frac{1}{x}\right)}$

$=-\frac{1}{2}$

(ix) $\lim_{n\to\infty}\frac{1}{n^2}(1+2+...+n)=\lim_{x\to\infty}\frac{n^2\left(1+\frac{1}{n}\right)}{2n^2}=\frac{1}{2}$

4. $\lim_{x\to 3}\frac{x^3-7x^2+15x-9}{x^4-5x^3+27x-27}$

**Ans.** 2/9

**Sol.** $\lim\limits_{x\to3}\dfrac{(x-3)\left(x^2-4x+3\right)}{\left(x^2-6x+9\right)\left(x^2+x-3\right)}=\lim\limits_{x\to3}\dfrac{(x-3)^2(x-1)}{(x-3)^2\left(x^2+x-3\right)}$

$=\dfrac{2}{9}$

5. $\lim\limits_{x\to2}\dfrac{\sqrt{1+\sqrt{2+x}}-\sqrt{3}}{x-2}$

**Ans.** $1/8\sqrt{3}$

**Sol.** $\lim\limits_{x\to2}\dfrac{\sqrt{2+x}-2}{\left(\sqrt{1+\sqrt{2+x}}+\sqrt{3}\right)(x-2)}=\lim\limits_{x\to2}\dfrac{(x-2)}{2\sqrt{3}\left(\sqrt{2+x}+2\right)(x-2)}$

$=\dfrac{1}{8\sqrt{3}}$

6. What can be said about $\lim\limits_{x\to\infty}\dfrac{a_0x^m+a_1x^{m-1}+....+a_m}{b_0x^n+b_1x^{n-1}+....+b_n}$, where $a_0, b_0\neq0$ and m, n are positive integers when

(i) $m>n$, (ii) $m=n$, (iii) $m<n$

$\lim\limits_{x\to\infty}\dfrac{a_0x^m+a_1x^{m-1}+....+a_m}{b_0x^n+b_1x^{n-1}+....+b_n}$ का मान ज्ञात कीजिए, जहाँ $a_0, b_0\neq0$ तथा m, n प्राकृत संख्याऐं है

जब (i) $m>n$, (ii) $m=n$, (iii) $m<n$

**Ans.** (i) D.N.E.; (ii) $\dfrac{a_0}{b_0}$ (iii) 0

**Sol.** $\lim\limits_{x\to\infty}\dfrac{a_0x^m+a_1x^{m-1}+....+a_m}{b_0x^n+b_1x^{n-1}+....+b_n}$

$\lim\limits_{x\to\infty}x^{m-n}\left(\dfrac{a_0+\dfrac{a_1}{x}+....+\dfrac{a_m}{x^m}}{b_0+\dfrac{b_1}{x}+....+\dfrac{b_n}{x^n}}\right)$

$$= \begin{cases} \text{DNE} & \text{if} \quad m > n \\ \dfrac{a_0}{b_0} & \text{if} \quad m = n \\ 0 & \text{if} \quad m < n \end{cases}$$

7. $\lim\limits_{n\to\infty} \dfrac{\sqrt{n^3-2n^2+1}+\sqrt[3]{n^4+1}}{\sqrt[4]{n^6+6n^5+2}-\sqrt[5]{n^7+3n^3+1}}$

**Ans.** **1**

**Sol.** $\ell\lim\limits_{n\to\infty} \dfrac{\sqrt{n^3-2n^2+1}+\sqrt[3]{n^4+1}}{\sqrt[4]{n^6+6n^5+2}-\sqrt[5]{n^7+3n^3+1}} = \ell\lim\limits_{n\to\infty} \dfrac{n^{\frac{3}{2}}\sqrt{1-\frac{2}{n}+\frac{1}{n^3}}+n^{\frac{4}{3}}\sqrt{1+\frac{1}{n^4}}}{n^{\frac{3}{2}}\sqrt{1+\frac{6}{n}+\frac{2}{n^6}}-n^{\frac{7}{5}}\sqrt{1+\frac{3}{n^4}+\frac{1}{n^7}}}$

$$= \ell\lim_{n\to\infty} \frac{\sqrt{1-\frac{2}{n}+\frac{1}{n^3}}+n^{-\frac{1}{6}}\sqrt{1+\frac{1}{n^4}}}{\sqrt{1+\frac{6}{n}+\frac{2}{n^6}}-n^{-\frac{1}{10}}\sqrt{1+\frac{3}{n^4}+\frac{1}{n^7}}} = \frac{1+0}{1-0} = 1$$

8. $\lim\limits_{x\to\infty} \dfrac{(x+1)^{10}+(x+2)^{10}+....+(x+100)^{10}}{x^{10}+10^{10}}$

**Ans.** **100**

**Sol.** $\lim\limits_{x\to\infty} \dfrac{\left(1+\frac{1}{x}\right)^{10}+.....+\left(1+\frac{100}{x}\right)^{10}}{1+\left(\frac{10}{x}\right)^{10}} = 100$

9. $\lim\limits_{x\to\pm\infty} \left(\sqrt{x^2-2x-1}-\sqrt{x^2-7x+3}\right)$

**Ans.** $x \to -\infty = -5/2, x \to \infty = 5/2$

**Sol.** $\lim\limits_{x\to\pm\infty} \dfrac{5x-4}{|x|\left(\sqrt{1-\frac{2}{x}-\frac{1}{x}}+\sqrt{1-\frac{7}{x}+\frac{3}{x^2}}\right)}$

$$= \begin{cases} \frac{5}{2} & \text{if} \quad x \to \infty \\ -\frac{5}{2} & \text{if} \quad x \to -\infty \end{cases}$$

10. $\lim\limits_{x\to\infty}\left(\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x}\right)$

**Ans.** 1/2

**Sol.** $\lim\limits_{x\to\infty}\left(\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x}\right)$

$= \lim\limits_{x\to\infty}\left(\frac{x+\sqrt{x+\sqrt{x}}-x}{\sqrt{x+\sqrt{x+\sqrt{x}}}+\sqrt{x}}\right)$ (By Rationalising)

$= \lim\limits_{x\to\infty}\left(\frac{\sqrt{1+\frac{\sqrt{x}}{x}}}{\sqrt{1+\sqrt{\frac{1}{x}+\sqrt{\frac{1}{x^3}}}}+1}\right) = \frac{1}{2}$

11. If $\lim\limits_{x\to-2}\frac{3x^2+\lambda x+2}{x^2+3x+2}$ exists and is equal to 'L' then the value of $(\lambda + L)$ is :

यदि $\lim\limits_{x\to-2}\frac{3x^2+\lambda x+2}{x^2+3x+2}$ विद्यमान है तथा 'L' के बराबर है, तो $(\lambda + L)$ का मान होगा :

**Ans.** **12**

**Sol.** $\lim\limits_{x\to-2}\frac{3x^2+\lambda x+2}{x^2+3x+2} = L$

$\therefore$ limit exist and $D^r \to 0$.

$\Rightarrow D^r \to 0$

$\Rightarrow 3(+4) - 2\lambda + \Rightarrow \lambda = 7$

$\therefore$ Given limit

$= \lim\limits_{x\to-2}\frac{3x^2+7x+2}{(x+2)(x+1)} = \lim\limits_{x\to-2}\frac{(3x+1)(x+2)}{(x+2)(x+1)}$

$$= \frac{-6+1}{-2+1} = 5$$

$$\therefore \lambda + L = 7 + 5 = 12$$

12. If limit of $f(x) = \begin{cases} x^3 + a & ; \quad x < -1 \\ bx + x & ; \quad -1 \le x < 1 \\ cx^2 & ; \quad x \ge 1 \end{cases}$ exist at $x = -1$ and $x = 1$ & both are equal, then $a + b + c$ =

यदि $f(x) = \begin{cases} x^3 + a & ; \quad x < -1 \\ bx + x & ; \quad -1 \le x < 1 \\ cx^2 & ; \quad x \ge 1 \end{cases}$ की सीमा (limit) $x = -1$ तथा $x = 1$ पर विद्यमान हो तथा दोनों समान हो, तो $a + b + c$ का मान होगा :

(A) 0 (B) 1 (C) –1 (D) 2

**Ans.** (A)

**Sol.** $-1 + a = -b - 1 = b + 1 = c$

$\Rightarrow b = -1, c = 0, a = 1$

13. If $P_n = \prod_{k=1}^{n}\left(1 + \frac{2}{k^2 + 3k}\right)$ then $\lim_{n\to\infty} P_n$ is equal to :

यदि $P_n = \prod_{k=1}^{n}\left(1 + \frac{2}{k^2 + 3k}\right)$ है, तो $\lim_{n\to\infty} P_n$ का मान होगा :

(A) 1 (B) 3 (C) 5 (D) 7

**Ans.** **B**

**Sol.** $P_n = \prod_{k=1}^{n}\left(\frac{k^2 + 3k + 2}{k^2 + 3k}\right) = \prod_{k=1}^{n}\left(\frac{k+1}{k}\right)\left(\frac{k+2}{k+3}\right)$

$$\lim_{x\to\infty} P_n = \frac{2\times 3}{1\times 4} \times \frac{3\times 4}{1\times 4} \times \frac{4\times 5}{3\times 6} \ldots\ldots \frac{(n+1)(n+2)}{n(n+3)} = 3$$

14. The value of $\lim_{x\to 2} \frac{\sqrt{x-2} + \sqrt{x} - \sqrt{2}}{\sqrt{x^2 - 4}}$ is equal to :

$\lim_{x\to 2}\dfrac{\sqrt{x-2}+\sqrt{x}-\sqrt{2}}{\sqrt{x^2-4}}$ का मान होगा :

(A) 1 (B) 2 (C) 1/2 (D) 1/4

**Ans. C**

**Sol.** $\lim_{x\to 2}\dfrac{\sqrt{x-2}+\sqrt{x}-\sqrt{2}}{\sqrt{(x-2)}\sqrt{(x+2)}}=\lim_{x\to 2}\left(\dfrac{1}{\sqrt{x+2}}+\dfrac{\sqrt{x-2}}{\left(\sqrt{x}+\sqrt{2}\right)\left(\sqrt{x+2}\right)}\right)$

$=\dfrac{1}{2}$

## ONE OR MORE THAN ONE CORRECT TYPE

15. Which of the following limit do not exist?

निम्न में से कौनसी सीमा विद्यमान नहीं होगी?

(A) $\lim_{x\to 1}\dfrac{x^3+x-2}{x^3-x^2-x+1}$ (B) $\lim_{x\to 2}\left[\dfrac{1}{x(x-2)^2}-\dfrac{1}{x^2-3x+2}\right]$

(C) $\lim_{x\to 1}\left[\dfrac{x+2}{x^2-5x+4}+\dfrac{x-4}{3(x^2-3x+2)}\right]$ (D) $\lim_{x\to\infty}\left(\dfrac{x^3}{2x^2-1}-\dfrac{x^2}{2x+1}\right)$

**Ans. AB**

**Sol.** (A) $\lim_{x\to 1}\dfrac{(x-1)\left(x^2+x+2\right)}{(x-1)^2(x+1)}=\text{DNE}$

(B) $\lim_{x\to 2}\left[\dfrac{1}{x(x-2)^2}-\dfrac{1}{(x-2)(x-1)}\right]=\lim_{x\to 2}\dfrac{-x(x-2)+x-1}{x(x-2)^2(x-1)}$

$=\lim_{x\to 2}\dfrac{-x^2+3x-1}{x(x-2)^2(x-1)}=\text{DNE}$

(C) $\lim_{x\to 1}\left[\dfrac{x+2}{(x-1)(x-4)}+\dfrac{x-4}{3(x-1)(x-2)}\right]$

$=\lim_{x\to 1}\dfrac{3\left(x^2-4\right)+(x-4)^2}{3(x-1)(x-2)(x-4)}=\lim_{x\to 1}\dfrac{4x^2-8x+4}{3(x-1)(x-2)(x-4)}$

$= 0$

(D) $\lim\limits_{x\to\infty}\left(\dfrac{2x^4+x^3-2x^4+x^2}{\left(2-\dfrac{1}{x^2}\right)\left(2+\dfrac{1}{x}\right)\cdot x^3}\right)=\dfrac{1}{4}$

16. Which of the following limit vanishes ?

निम्न में से कौनसी सीमा विलुप्त (मान शून्य है) होगी?

(A) $\lim\limits_{x\to\infty}\dfrac{\sin x}{x}$

(B) $\lim\limits_{x\to\infty}\dfrac{\text{arc}\tan x}{x}$

(C) $\lim\limits_{x\to\infty}\dfrac{x+\sin x}{x+\cos x}$

(D) $\lim\limits_{x\to1}\dfrac{\text{arc}\sin x}{\tan\dfrac{\pi x}{2}}$

**Ans. ABD**

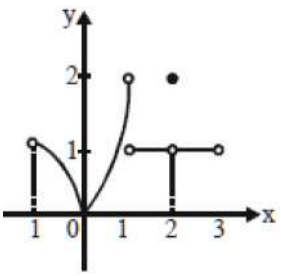
**Sol.** (A) $\lim\limits_{x\to\infty}\dfrac{\sin x}{x}=0$

(B) $\lim\limits_{x\to\infty}\dfrac{\tan^{-1}x}{x}=0$

(C) $\lim\limits_{x\to\infty}\dfrac{x+\sin x}{x+\cos x}=1$

(D) $\lim\limits_{x\to1}\dfrac{\sin^{-1}x}{\tan\dfrac{\pi x}{2}}=0$

17. Which of the following statement are true for the function f defined for $-1\le x\le 3$ in the figure shown :

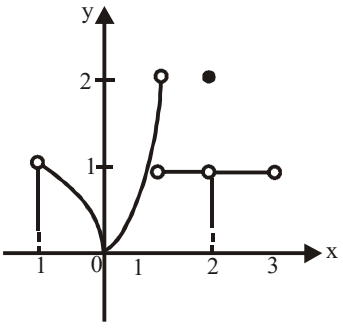



(A) $\lim\limits_{x\to-1^+}f(x)=1$

(B) $\lim\limits_{x\to2}f(x)$ does not exist

(C) $\lim\limits_{x\to 1^-} f(x) = 2$ (D) $\lim\limits_{x\to 0^+} f(x) = \lim\limits_{x\to 0^-} f(x)$

फलन f अन्तराल $-1 \le x \le 3$ में परिभाषित है। निम्न में से कौनसे कथन सत्य है :




(A) $\lim\limits_{x\to -1^+} f(x) = 1$ (B) $\lim\limits_{x\to 2} f(x)$ विद्यमान नहीं है

(C) $\lim\limits_{x\to 1^-} f(x) = 2$ (D) $\lim\limits_{x\to 0^+} f(x) = \lim\limits_{x\to 0^-} f(x)$

**Ans. ACD**

**Sol.** (A) $\lim\limits_{x\to -1^+} f(x) = \lim\limits_{h\to 0} f(-1+h) = 1$

$\lim\limits_{x\to 1^+} f(x) = \lim\limits_{h\to 0} f(1-h) = 2$

(B) $\lim\limits_{h\to 0} f(2+h) = 1 = \lim\limits_{h\to 0} f(2-h) = 1$

(C) $\lim\limits_{h\to 0} f(0+h) = \lim\limits_{h\to 0} f(0-h)$

18. Let $f(x) = x + \sqrt{x^2+2x}$ and $g(x) = \sqrt{x^2+2x} - x$, then :

माना $f(x) = x + \sqrt{x^2+2x}$ तथा $g(x) = \sqrt{x^2+2x} - x$ हो, तो :

(A) $\lim\limits_{x\to\infty} g(x) = 1$ (B) $\lim\limits_{x\to\infty} f(x) = 1$ (C) $\lim\limits_{x\to -\infty} f(x) = -1$ (D) $\lim\limits_{x\to -\infty} g(x) = -1$

**Ans. AC**

**Sol.** $\lim\limits_{x\to\infty} g(x) = \lim\limits_{x\to\infty} \dfrac{2x}{\sqrt{1+\dfrac{2}{x}}|x| + x}$

$= \lim\limits_{x\to\infty} \dfrac{2}{\sqrt{1+\dfrac{2}{x}}+1} = 1$

$$\lim_{x\to-\infty} g(x) = \lim_{x\to-\infty} \frac{2}{1-\sqrt{1+\frac{2}{x}}} = \text{DNE}$$

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} x + \sqrt{x^2+2x} = \text{DNE}$$

$$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{-2x}{x-\sqrt{x^2-2x}} = \lim_{x\to-\infty} \frac{-2}{1+\sqrt{1-\frac{2}{x}}} = -1$$

1. Calculate the following limits :

निम्न सीमाऐं ज्ञात कीजिए :

(i) $\lim\limits_{x\to 0}\dfrac{1-\cos^3 x}{x\sin 2x}$ (ii) $\lim\limits_{x\to 0}\left(\dfrac{1}{\sin x}-\dfrac{1}{\tan x}\right)$ (iii) $\lim\limits_{x\to 0}\dfrac{\sin 4x}{1-\sqrt{1-x}}$ (iv) $\lim\limits_{x\to \pi/2}\dfrac{\cot x-\cos x}{(\pi-2x)^3}$

(v) $\lim\limits_{\alpha\to\beta}\dfrac{\sin^2\alpha-\sin^2\beta}{\alpha^2-\beta^2}$ (vi) $\lim\limits_{x\to\infty}\left(\cos\sqrt{x+1}-\cos\sqrt{x}\right)$ (vii) $\lim\limits_{x\to\pi/2}\dfrac{1+\cos 2x}{(\pi-2x)^2}$

(viii) $\lim\limits_{x\to\pi/2}\dfrac{\cos 3x+3\cos x}{\left(\dfrac{\pi}{2}-x\right)^3}$ (ix) $\lim\limits_{x\to\frac{\pi}{6}}\dfrac{2-\sqrt{3}\cos x-\sin x}{(6x-\pi)^2}$ (x) $\lim\limits_{x\to\pi}\dfrac{1-\sin\dfrac{x}{2}}{\cos\dfrac{x}{2}\left(\cos\dfrac{x}{4}-\sin\dfrac{x}{4}\right)}$

**Ans.** (i) 3/4 (ii) 0 (iii) 8 (iv) 1/16 (v) $\left(\dfrac{\sin 2\beta}{2\beta}\right)$ (vi) 0 (vii) 1/2 (viii) 4 (ix) 1/36 (x) $1/\sqrt{2}$

**Sol.** (i) $\lim\limits_{x\to 0}\dfrac{1-\cos^3 x}{2x^2}=\lim\limits_{x\to 0}\dfrac{(1-\cos x)(1+\cos^2 x+2\cos x)}{2x^2}$

$=\dfrac{3}{2}\lim\limits_{x\to 0}\dfrac{(1-\cos x)}{x^2}=\dfrac{3}{4}$

(ii) $\lim\limits_{x\to 0}\dfrac{(1-\cos x)}{\sin x}=\lim\limits_{x\to 0}\dfrac{2\sin^2\dfrac{x}{2}}{\dfrac{x^2}{4}}\dfrac{x^2}{4\cdot\sin x}=0$

(iii) $\lim\limits_{x\to 0}\dfrac{\sin 4x}{4x}\cdot 4x\cdot\dfrac{\left(1+\sqrt{1-x}\right)}{x}=8$

(iv) $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{\cos x(1-\sin x)}{8\sin x\left(\dfrac{\pi}{2}-x\right)^3}=\lim\limits_{h\to 0}\dfrac{+\sinh(1-\cosh)}{8\cosh h^3}$

$$= \lim_{n\to 0} \frac{2\sin^2 \frac{h}{2}}{8\cosh \cdot 4\frac{h^2}{4}} = \frac{1}{16}$$

(v) $\lim_{\alpha\to\beta} \frac{\sin(\alpha+\beta)}{\alpha+\beta} \frac{\sin(\alpha-\beta)}{\alpha-\beta} = \frac{\sin 2\beta}{2\beta}$

(vi) $\lim_{x\to\infty} \left( 2\sin \frac{\sqrt{x+1}+\sqrt{x}}{2} \cdot \sin\left( \frac{\sqrt{x+1}-\sqrt{x}}{2} \right) \right)$

$$= 2\lim_{x\to\infty} \frac{\sin\left( \frac{\sqrt{x+1}+\sqrt{x}}{2} \right)}{\in[-1,1]} \cdot \sin\left( \frac{1}{2\left(\sqrt{x+1}+\sqrt{x}\right)} \right) = 0$$

(vii) $\lim_{x\to\frac{\pi}{2}} \frac{2\cos^2 x}{4\left(\frac{\pi}{2}-x\right)^2} = \lim_{n\to 0} \frac{2\sin^2 h}{4h^2} = \frac{1}{2}$

(viii) $\lim_{x\to\frac{\pi}{2}} \frac{-3\cos x + 4\cos^3 x + 3\cos x}{\left(\frac{\pi}{2}-x\right)^3}$

$$\lim_{n\to 0} \frac{4\sin^3 h}{4^3} = 4$$

(ix) $\lim_{x\to\frac{\pi}{6}} \frac{2\left[1-\left(\cos\left(x-\frac{\pi}{6}\right)\right)\right]}{36\left(x-\frac{\pi}{6}\right)^2}$

$$= \lim_{h\to 0} \frac{4\sin^2 \frac{h}{2}}{36h^2} = \frac{1}{36}$$

(x) $\lim\limits_{x\to\pi}\dfrac{1-\sin\left(\dfrac{x}{2}\right)}{\cos\dfrac{x}{2}\left(\dfrac{\cos x}{4}-\dfrac{\sin x}{4}\right)}$

$$=\lim_{h\to0}\frac{1-\cos\frac{h}{2}}{-\sin\left(\frac{h}{2}\right)\left(\cos\left(\frac{\pi}{4}+\frac{h}{4}\right)-\sin\left(\frac{\pi}{4}+\frac{h}{4}\right)\right)}$$

$$=\lim_{h\to0}\frac{2\sin^2\frac{h}{2}}{-\sin\frac{h}{2}}\frac{\cos\left(\frac{\pi}{4}+\frac{h}{4}\right)+\sin\left(\frac{\pi}{4}+\frac{h}{4}\right)}{\cos\left(\frac{\pi}{2}+\frac{h}{2}\right)}$$

$$=\lim_{h\to0}\frac{+4}{\sqrt{2}}\cdot\frac{\sin^2\frac{h}{4}}{\frac{h^2}{16}}\cdot\frac{\frac{h^2}{16}}{\sin^2\frac{h}{2}}=\frac{4}{4\sqrt{2}}=\frac{1}{\sqrt{2}}$$

2. $\lim\limits_{x\to1}\dfrac{\sqrt[3]{7+x^3}-\sqrt{3+x^2}}{x-1}$

**Ans.** –1/4

**Sol.** $\lim\limits_{x\to1}\dfrac{\left(7+x^3\right)^{\frac{1}{3}}-\left(3+x^2\right)^{\frac{1}{2}}}{x-1}$

$$=\lim_{h\to0}\frac{\left(7+(1+h)^3\right)^{\frac{1}{3}}-\left(3+(1+h)^2\right)^{\frac{1}{2}}}{h}$$

$$=2\lim_{h\to0}\frac{\left(\left(1+\frac{h^3+3h^2+3h}{8}\right)^{\frac{1}{3}}-\left(1+\frac{h^2+2h}{4}\right)^{\frac{1}{2}}\right)}{h}$$

$$=2\lim_{h\to0}\frac{1}{h}\left[1+\frac{1}{3}\left(\frac{h^3+3h^2+3h}{8}\right)+....\left(1+\frac{1}{2}\left(\frac{h^2+2h}{4}\right)\right)+.....\right]$$

$$= 2\cdot\left(\frac{1}{8}-\frac{1}{4}\right) = -\frac{1}{4}$$

3. $\lim\limits_{x\to 1}\dfrac{(x+x^2+x^3+....+x^n)-n}{x-1}$

**Ans.** $\dfrac{n(n+1)}{2}$

**Sol.** $\lim\limits_{x\to 1}\dfrac{\left(x+x^2+....+x^n\right)-n}{x-1}$

$$= \lim_{x\to 1}\frac{(x-1)+\left(x^2-1\right)+.....+\left(x^n-1\right)}{(x-1)}$$

$$= \lim_{x\to 1} 1+(x+1)+\left(x^2+x+1\right)+.....+\left(x^{n-1}+x^{n-2}+....+1\right)$$

$$= 1 + 2 + 3 + 4 + ..... + n$$

$$= \frac{n(n+1)}{2}$$

4. $\lim\limits_{x\to\frac{\pi}{2}}\left(2x\tan x-\dfrac{\pi}{\cos x}\right)$

**Ans.** –2

**Sol.** $\lim\limits_{x\to\frac{\pi}{2}}\left(2x\tan x-\dfrac{\pi}{\cos x}\right) = \lim\limits_{x\to\frac{\pi}{2}}\dfrac{2x\sin x-\pi}{\cos x}$

$$= \lim_{h\to 0}\frac{2\left(\frac{\pi}{2}+h\right)\sin\left(\frac{\pi}{2}+h\right)-\pi}{\cos\left(\frac{\pi}{2}+h\right)}$$

$$= \lim_{h\to 0}\frac{(\pi+2h)\cosh-\pi}{-\sinh} = \lim_{h\to 0}\frac{\pi(\cosh-1)+2h\cosh}{-h}$$

$$= \lim_{h\to 0}\frac{-2\pi\sin^2\frac{h}{2}+2h\cosh}{-h}$$

$$= \lim_{h\to 0}\left(-2\pi\cdot\frac{\sin^2\frac{h}{2}}{\frac{h^2}{4}}\cdot\frac{h^2}{-4\cdot h}+\frac{2\cosh}{-1}\right)$$

$$= -2$$

5. $\lim\limits_{x\to\frac{\pi}{2}} \tan^2 x\left(\sqrt{2\sin^2 x+3\sin x+4}-\sqrt{\sin^2 x+6\sin x+2}\right)$

**Ans.** $\frac{1}{12}$

**Sol.** $\lim\limits_{x\to\frac{\pi}{2}} \tan^2 x\left(\sqrt{2\sin^2 x+3\sin x+4}-\sqrt{\sin^2 x+6\sin x+2}\right)$

$$= \lim_{x\to\frac{\pi}{2}} \tan^2 x\left(\frac{2\sin^2 x+3\sin x+4-(\sin^2 x+6\sin x+2)}{\sqrt{2\sin^2 x+3\sin x+4}+\sqrt{\sin^2 x+6\sin x+2}}\right) = \lim_{x\to\frac{\pi}{2}} \tan^2 x\cdot\frac{(\sin^2 x-3\sin x+2)}{\sqrt{2+3+4}+\sqrt{1+6+2}}$$

$$= \lim_{x\to\frac{\pi}{2}} \frac{1}{6}\left[\frac{\sin^2 x-3\sin x+2}{\cos^2 x}\right] \quad \left(\frac{0}{0}\text{ form}\right)\left(\frac{0}{0}\text{ रूप}\right) \quad \text{(use L'Hospital rule)} \quad \text{(L'Hospital नियम से)}$$

$$= \frac{1}{6}\lim_{x\to\frac{\pi}{2}} \frac{2\sin x\cos x-3\cos x}{2\cos x(-\sin x)} = \frac{1}{6}\lim_{x\to\frac{\pi}{2}} \frac{2\sin x-3}{-2\sin x} = \left(\frac{1}{6}\right)\left(\frac{1}{2}\right) = \frac{1}{12}$$

6. $\lim\limits_{n\to\infty}\left(\cos\frac{x}{2}.\cos\frac{x}{4}...\cos\frac{x}{2^n}\right)$

**Ans.** $\frac{\sin x}{x}$

**Sol.** $\lim\limits_{n\to\infty} \cos\frac{x}{2}\cos\frac{x}{2^2}\cos\frac{x}{2^3}\cos\frac{x}{2^4}.....\cos\frac{x}{2^n}$

$$= \lim_{n\to\infty} \frac{\sin x}{2^n\sin\frac{x}{2^n}} = \lim_{n\to\infty} \frac{\sin x}{x}\cdot\left(\frac{\frac{x}{2^n}}{\sin\frac{x}{2^n}}\right) \qquad \left(\because \lim_{n\to\infty} \frac{x}{2^n}=0\right)$$

$$= \frac{\sin x}{x}$$

7. $\lim\limits_{x\to\infty} x\left(\arctan\dfrac{x+1}{x+2}-\arctan\dfrac{x}{x+2}\right)$

**Ans.** $\dfrac{1}{2}$

**Sol.** $\lim\limits_{x\to\infty} x\tan^{-1}\left(\dfrac{\dfrac{1}{x+2}}{1+\dfrac{x(x+1)}{(x+2)^2}}\right)=\lim\limits_{x\to\infty} x\tan^{-1}\left(\dfrac{x+2}{(x+2)^2+x(x+1)}\right)$

$$=\lim_{x\to\infty}\frac{\tan^{-1}\left(\dfrac{x+2}{2x^2+5x+4}\right)}{\dfrac{1}{x}}$$

$$=\lim_{x\to\infty}\frac{x\cdot\tan^{-1}\left(\dfrac{x+2}{2x^2+5x+4}\right)}{\dfrac{x+2}{2x^2+5x+4}}\cdot\frac{x+2}{2x^2+5x+4}$$

$$=\lim_{x\to\infty}\frac{x^2+2x}{2x^2+5x+4}=\frac{1}{2}$$

8. $\lim\limits_{x\to0}\dfrac{1-\cos x\sqrt{\cos 2x}}{x^2}$

**Ans.** $\dfrac{3}{2}$

**Sol.** $\lim\limits_{x\to0}\dfrac{1-\cos x\sqrt{\cos 2x}}{x^2}$

$$=\lim_{x\to0}\frac{1-\cos^2 x(\cos 2x)}{x^2\left(1+\cos x\sqrt{\cos 2x}\right)}$$

$$=\lim_{x\to0}\frac{\sin^2 x+\cos^2 x(1-\cos 2x)}{x^2\left(1+\cos x\sqrt{\cos 2x}\right)}$$

$$= \lim_{x\to 0} \frac{\sin^2 x\left(1+2\cos^2 x\right)}{x^2\left(1+\cos x\sqrt{\cos 2x}\right)} = \frac{3}{2}$$

9. $\lim\limits_{\theta\to\frac{\pi}{4}} \dfrac{\sqrt{2}-\cos\theta-\sin\theta}{(4\theta-\pi)^2}$.

**Ans.** $\dfrac{1}{16\sqrt{2}}$

**Sol.** $\lim\limits_{\theta\to\frac{\pi}{4}} \dfrac{\sqrt{2}-\cos\theta-\sin\theta}{(4\theta-\pi)^2} = \lim\limits_{\theta\to\frac{\pi}{4}} \dfrac{\sqrt{2}-\sqrt{2}\left[\cos\left(\theta-\frac{\pi}{4}\right)\right]}{(4\theta-\pi)^2}$

$$= \lim_{\theta\to\frac{\pi}{4}} \frac{\sqrt{2}\left[1-\cos\left(\theta-\frac{\pi}{4}\right)\right]}{16\left(\theta-\frac{\pi}{4}\right)^2} = \frac{1}{16\sqrt{2}}.$$

10. Let $P_n = \prod\limits_{k=2}^{n}\left(1-\dfrac{1}{{}^{k+1}C_2}\right)$. If $\lim\limits_{n\to\infty} P_n$ can be expressed as lowest rational in the form $\dfrac{a}{b}$, find the value of (a + b).

माना $P_n = \prod\limits_{k=2}^{n}\left(1-\dfrac{1}{{}^{k+1}C_2}\right)$ यदि $\lim\limits_{n\to\infty} P_n$ को $\dfrac{a}{b}$ रूप (सरलतम) में व्यक्त किया जा सकता है, तो (a + b) का मान ज्ञात कीजिए।

**Ans.** 4

**Sol.** $P_n = \prod\limits_{k=2}^{n}\left(1-\dfrac{1}{{}^{k+1}C_2}\right)$

$$P_n = \prod_{k=2}^{n}\left(1-\frac{|2|k-1}{\underline{|k+1}}\right)$$

$$P_n = \prod_{k=2}^{n}\left(\frac{(k+1)k-2}{(k+1)k}\right)$$

$$P_n = \prod_{k=2}^{n} \frac{k^2+k-2}{k(k+1)}$$

$$P_n = \prod_{k=2}^{n} \frac{(k-1)(k+2)}{k(k+1)}$$

$$\Rightarrow P_n = \frac{n+2}{3n}$$

$$\Rightarrow \lim_{n\to\infty} P_n = \frac{1}{3}$$

11. Let $a = \min\{x^2+2x+3, x \in R\}$ and $b = \lim_{\theta\to 0} \frac{1-\cos\theta}{\theta^2}$. Then value of $\sum_{r=0}^{n} a^r.b^{n-r}$ is :

(A) $\frac{2^{n+1}-1}{3.2^n}$ (B) $\frac{2^{n+1}+1}{3.2^n}$ (C) $\frac{4^{n+1}-1}{3.2^n}$ (D) None of these

माना $a = \min\{x^2+2x+3, x \in R\}$ तथा $b = \lim_{\theta\to 0} \frac{1-\cos\theta}{\theta^2}$ तो $\sum_{r=0}^{n} a^r.b^{n-r}$ का मान होगा :

(A) $\frac{2^{n+1}-1}{3.2^n}$ (B) $\frac{2^{n+1}+1}{3.2^n}$ (C) $\frac{4^{n+1}-1}{3.2^n}$ (D) इनमें से कोई नहीं

**Ans.** **C**

**Sol.** $a = 2, \quad b = \lim_{\theta\to 0} \frac{1-\cos\theta}{\theta^2} = \frac{1}{2}$

$$\sum_{r=0}^{n} a^r b^{n-r} = b^n + ab^{n-1} + .... + a^n$$

$$= \frac{b^n\left(1-\left(\frac{a}{b}\right)^{n+1}\right)}{1-\frac{a}{b}}$$

$$= \left(\frac{b^{n+1}-a^{n+1}}{b-a}\right)\cdot b^{-1} = \frac{\left(\left(\frac{b}{a}\right)^{n+1}-1\right)}{\left(\frac{b}{a}-1\right)}\cdot a^{n+1}$$

$$= \left( \frac{1-\left(\frac{a}{b}\right)^{n+1}}{1-\frac{a}{b}} \right) \cdot b^n$$

$$= \frac{4^{n+1}-1}{3 \cdot 2^n}$$

12. If $\lim\limits_{n\to\infty}\left(\sqrt{2n^2+n}-\lambda\sqrt{2n^2-n}\right)=\frac{1}{\sqrt{2}}$ (where $\lambda$ is a real number), then :

यदि $\lim\limits_{n\to\infty}\left(\sqrt{2n^2+n}-\lambda\sqrt{2n^2-n}\right)=\frac{1}{\sqrt{2}}$ (जहाँ $\lambda$ वास्तविक संख्या है), तो :

(A) $\lambda = 1$ (B) $\lambda = -1$

(C) $\lambda = \pm 1$ (D) $\lambda \in (-\infty, 1)$

**Ans. A**

**Sol.** $$\lim_{n\to\infty} \frac{2n^2+n-\lambda^2\left(2n^2-n\right)}{\sqrt{2n^2+n}+\lambda\sqrt{2n^2-n}}$$

$$= \lim_{n\to\infty} \frac{(1-\lambda)2n^2+\left(1+\lambda^2\right)n}{n\left(\sqrt{2+\frac{1}{n}}+\lambda\sqrt{2-\frac{1}{n}}\right)}$$

for exist of limit coefficients of $n^2$ should be zero.

so $\lambda = 1$

13. If $\lim\limits_{x\to a}\frac{2x-\sqrt{x^2+3a^2}}{\sqrt{x+a}-\sqrt{2a}}=\sqrt{2}$ (w here $a \in R^+$), then a is equal to :

यदि $\lim\limits_{x\to a}\frac{2x-\sqrt{x^2+3a^2}}{\sqrt{x+a}-\sqrt{2a}}=\sqrt{2}$ (जहाँ $a \in R^+$), तो a का मान होगा :

(A) $\frac{1}{3}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{9}$

**Ans. D**

**Sol.** $\lim\limits_{n\to a}\dfrac{\left(4x^2-\left(x^2+3a^2\right)\right)\left(\sqrt{x+a}+\sqrt{2a}\right)}{\left((x+a)-2a\right)\left(2x+\sqrt{x^2+3a^2}\right)}$

$\Rightarrow \lim\limits_{n\to a}\dfrac{3\left(x^2-a^2\right)}{(x-a)}\dfrac{2\sqrt{2a}}{4a}=\sqrt{2}$

$\Rightarrow \dfrac{3(2a).2\sqrt{2a}}{4a}=\sqrt{2}\Rightarrow a=\dfrac{1}{9}$

14. Let $U_n=\dfrac{n!}{(n+2)!}$ where $n\in N$. If $S_n=\sum\limits_{n=1}^{n}U_n$ then $\lim\limits_{n\to\infty}S_n$ equals :

(A) 2 (B) 1

(C) 1/2 (D) Non existent

माना $U_n=\dfrac{n!}{(n+2)!}$ जहाँ $n\in N$. यदि $S_n=\sum\limits_{n=1}^{n}U_n$ तो $\lim\limits_{n\to\infty}S_n$ का मान होगा :

(A) 2 (B) 1

(C) 1/2 (D) विद्यमान नही है

**Ans.** **C**

**Sol.** $U_n=\dfrac{\lfloor n}{\lfloor n+2}$

$S_n=\sum\limits_{r=1}^{n}U_r=\sum\limits_{r=1}^{n}\dfrac{1}{(r+2)(r+1)}$

$=\sum\limits_{r=1}^{n}\left[\dfrac{1}{r+1}-\dfrac{1}{r+2}\right]=\dfrac{1}{2}-\dfrac{1}{3}+\dfrac{1}{3}-\dfrac{1}{4}+.....+\dfrac{1}{n}-\dfrac{1}{n+1}+\dfrac{1}{n+1}-\dfrac{1}{n+2}$

$\Rightarrow \lim\limits_{n\to\infty}S_n=\dfrac{1}{2}$

15. $\lim\limits_{x\to 1/\sqrt{2}}\dfrac{x-\cos(\sin^{-1}x)}{1-\tan(\sin^{-1}x)}=$

(A) $\dfrac{1}{\sqrt{2}}$ (B) $-\dfrac{1}{\sqrt{2}}$

(C) $\sqrt{2}$ (D) $-\sqrt{2}$

**Ans. B**

**Sol.** $\lim\limits_{x\to\frac{1}{\sqrt{2}}} \dfrac{x-\cos\left(\sin^{-1}x\right)}{1-\tan\left(\sin^{-1}x\right)} \left(\dfrac{0}{0} \text{ form}\right)$

let $\sin^{-1}x = \theta$

$\Rightarrow x = \sin\theta$

$= \lim\limits_{x\to\frac{\pi}{4}} \dfrac{\sin\theta-\cos\theta}{1-\tan\theta}$

$= \lim\limits_{x\to\frac{\pi}{4}} \left(-\cos\theta\right) = -\dfrac{1}{\sqrt{2}}$

16. $\lim\limits_{x\to 0} \dfrac{l\,n(\sin 3x)}{l\,n(\sin x)}$ is equal to :

(A) 0 (B) 1

(C) 2 (D) Non existent

$\lim\limits_{x\to 0} \dfrac{l\,n(\sin 3x)}{l\,n(\sin x)}$ का मान है :

(A) 0 (B) 1

(C) 2 (D) विद्यमान नहीं है

**Ans. B**

**Sol.** $\lim\limits_{x\to 0} \dfrac{\ln\left(\sin 3x\right)}{\ln\left(\sin x\right)} = \lim\limits_{x\to 0} \dfrac{\ln\sin x + \ln\left(3-4\sin^2 x\right)}{\ln\sin x} = 1$

17. $\underset{x\to 0}{\text{Limit}}\ \sin^{-1}(\sec x)$ :

(A) is equal to $\pi/2$ (B) is equal to 1

(C) is equal to zero (D) none of these

$\underset{x\to 0}{\text{Limit}}\ \sin^{-1}(\sec x)$ का मान है :

(A) $\pi/2$ (B) 1

(C) शून्य (D) इनमें से कोई नहीं

**Ans.** D

**Sol.** $\lim\limits_{x\to 0^+} \sec x > 1$ So limit not exist

18. The value of $\lim\limits_{x\to 0}\dfrac{x^8}{1-\cos(1-\cos(1-\cos x))}$ is equal to :

$\lim\limits_{x\to 0}\dfrac{x^8}{1-\cos(1-\cos(1-\cos x))}$ का मान होगा :

(A) 32 (B) 64 (C) 128 (D) 256

**Ans.** C

**Sol.** $\lim\limits_{x\to 0}\dfrac{\left(1-\cos\left(1-\cos x\right)\right)^2}{1-\cos\left(1-\cos\left(1-\cos x\right)\right)}\times\dfrac{\left(1-\cos x\right)^4}{\left(1-\cos\left(1-\cos x\right)\right)^2}\times\dfrac{x^8}{\left(1-\cos x\right)^4}$

$= 2 \times 2^2 \times 2^4 = 2^7 = 128$

19. The value of $\lim\limits_{x\to\infty} x\left[\tan^{-1}\left(\dfrac{2x+1}{2x+4}\right)-\dfrac{\pi}{4}\right]$ is equal to :

$\lim\limits_{x\to\infty} x\left[\tan^{-1}\left(\dfrac{2x+1}{2x+4}\right)-\dfrac{\pi}{4}\right]$ का मान होगा :

(A) $\dfrac{1}{2}$ (B) $-\dfrac{1}{2}$ (C) $\dfrac{3}{4}$ (D) $-\dfrac{3}{4}$

**Ans.** D

**Sol.** Let $\dfrac{2x+1}{2x+4} = \tan t \Rightarrow t \to \dfrac{\pi}{4}$

and $x = \dfrac{4\tan t - 1}{2 - 2\tan t}$

$\therefore \lim\limits_{x\to\infty} x\left(\tan^{-1}\left(\dfrac{2x+1}{2x+4}\right)-\dfrac{\pi}{4}\right)$

$$\Rightarrow \lim_{t\to\frac{\pi}{4}} \frac{(4\tan t-1)}{2-2\tan t}\left(t-\frac{\pi}{4}\right)$$

$$\Rightarrow \lim_{t\to\frac{\pi}{4}} \frac{4\sec^2 t\left(t-\frac{\pi}{4}\right)+(4\tan-1)}{-2\sec^2 t} = -\frac{3}{4}$$

20. The value of $\lim\limits_{x\to\frac{1}{\sqrt{2}}} \dfrac{2\sin^{-1}x-\frac{\pi}{2}}{1-2x^2}$ is equal to :

$\lim\limits_{x\to\frac{1}{\sqrt{2}}} \dfrac{2\sin^{-1}x-\frac{\pi}{2}}{1-2x^2}$ का मान होगा :

(A) $-1$ (B) 0 (C) 1 (D) D.N.E.

**Ans.** **A**

**Sol.** put $x=\sin\theta$

$$\Rightarrow \lim_{\theta\to\frac{\pi}{4}} \frac{2\theta-\frac{\pi}{2}}{1-2\sin^2\theta} = \lim_{\theta\to\frac{\pi}{4}} \frac{2\theta-\frac{\pi}{2}}{1-2\sin^2\theta}$$

$$\lim_{\theta\to\frac{\pi}{4}} \frac{\left(2\theta-\frac{\pi}{2}\right)}{\sin\left(\frac{\pi}{2}-2\theta\right)} = -1$$

1. Calculate the following limits :

निम्न सीमाऐं ज्ञात कीजिए :

(i) $\lim_{n\to\infty} n\sin\left(\frac{\pi}{4n}\right)\cos\left(\frac{\pi}{4n}\right)$ (ii) $\lim_{x\to0}\frac{\sin x^o}{x}$ (iii) $\lim_{x\to0}\frac{e^{3x}-e^{4x}}{x}$ (iv) $\lim_{x\to\infty}\frac{x+\sin x}{x+\cos x}$

(v) $\lim_{x\to0}\frac{\sqrt{(1-\cos 2x)/2}}{x}$ (vi) $\lim_{x\to0}\frac{x(5^x-1)}{1-\cos x}$ (vii) $\lim_{x\to0}\frac{a^x+b^x+c^x-3}{x}$ (viii) $\lim_{x\to2}\frac{\sin(e^{x-2}-1)}{\log(x-1)}$

(ix) $\lim_{x\to\infty} x(e^{1/x}-1)$ (x) $\lim_{x\to\infty}\frac{a^x}{a^x+1}(a>0)$ (xi) $\lim_{x\to\infty}\frac{a^x-a^{-x}}{a^x+a^{-x}}(a>0)$ (xii) $\lim_{x\to e}\frac{\ln x-1}{x-e}$

**Ans.** (i) $\pi/4$ (ii) $\pi/180$ (iii) $-1$ (iv) 1

(v) D.N.E. (vi) $2\ln 5$ (vii) $\ln(abc)$ (viii) 1

(ix) 1 (x) 0 if $0<a<1$, 1/2 if $a=1$, 1 if $a>1$

(xi) $-1$ if $0<a<1$, 0 if $a=1$, 1 if $a>1$

(xii) $e^{-1}$

**Sol.** (i) $\lim_{n\to\infty} n\sin\left(\frac{\pi}{4n}\right)\cos\left(\frac{\pi}{4n}\right)=\frac{\pi}{4}$

(ii) $\lim_{x\to0}\frac{\sin\left(\frac{\pi x}{180}\right)}{x}=\frac{\pi}{180}$

(iii) $\lim_{x\to0}\frac{e^{3x}-e^{4x}}{x}=-1$

(iv) $\lim_{x\to\infty}\frac{x+\sin x}{x+\cos x}=1$

(v) $\lim_{x\to0}\frac{\sqrt{1-\frac{\cos 2x}{2}}}{x}=\lim_{x\to0}\frac{|\sin x|}{x}=\text{DNE}$

(vi) $\lim_{x\to 0}\frac{x\left(5^x-1\right)}{1-\cos x}=\lim_{x\to 0}\frac{\left(e^{x\log 5}-1\right)}{\frac{x}{2}}$

$=2\log_e 5$

(vii) $\lim_{x\to 0}\frac{a^x+b^x+c^x-3}{x}=\lim_{x\to 0}\frac{e^{x\ln a}+e^{x\ln b}+e^{x\ln c}-3}{x}$

$=\ln(abc)$

(viii) $\lim_{x\to 2}\frac{\sin\left(e^{x-2}-1\right)}{\log\left(x-1\right)}=\lim_{x\to 2}\frac{e^{x-2}-1}{x-2}=1$

(ix) $\lim_{x\to\infty}x\left(e^{\frac{1}{x}}-1\right)=\lim_{y\to 0}\frac{e^y-1}{y}=1$

(x) $\lim_{x\to\infty}\frac{a^x}{a^x+1}\left(a>0\right)$

C–I : If $0<a<1$

then $\lim_{x\to\infty}\frac{a^x}{a^x+1}=0$

C–II : If $a=1$

then $\lim_{x\to\infty}\frac{a^x}{a^x+1}=\frac{1}{2}$

C–III : If $a>1$

then $\lim_{x\to\infty}\frac{a^x}{a^x+1}=\lim_{x\to\infty}1-\frac{1}{a^x+1}=1$

(xi) $\lim_{x\to\infty}\frac{a^{2x}-1}{a^{2x}+1}=\begin{cases}-1 & 0<a<1\\ 0 & a=1\\ 1 & a>1\end{cases}$

(xii) $\lim_{x\to e}\frac{\ln\left(\frac{x}{e}\right)}{x-e}=\lim_{x\to e}\frac{\ln\left(1+\frac{x}{e}-1\right)}{e\left(\frac{x}{e}-1\right)}$

$=\frac{1}{e}$

2. $\lim_{x\to 2}\frac{(\cos\alpha)^x-(\sin\alpha)^x-\cos 2\alpha}{x-2}$ where $0<\alpha<\pi/2$.

$\lim_{x\to 2}\frac{(\cos\alpha)^x-(\sin\alpha)^x-\cos 2\alpha}{x-2}$ जहाँ $0<\alpha<\pi/2$.

**Ans.** $(\cos^2\alpha.\ell n\cos\alpha-\sin^2\alpha.\ell n\sin\alpha)$

**Sol.** $\lim_{x\to 2}\frac{(\cos\alpha)^x+(\sin\alpha)^x-1}{x-2}$

$=\lim_{h\to 0}\frac{\cos^2\alpha\ \cos^h\alpha+\sin^2\alpha\ \sin^h\alpha-1}{h}$

$=\lim_{h\to 0}\frac{\cos^h\alpha-\sin^2\alpha\cos^h\alpha+\sin^2\alpha\sin^h\alpha-1}{h}$

$=\lim_{h\to 0}\frac{\cos^h\alpha-1}{h}+\sin^2\alpha\left(\frac{\sin^h\alpha-1-\cos^h\alpha+1}{h}\right)$

$=\ell n\cos\alpha+\sin^2\alpha\,(\ell n\sin\alpha-\ell n\cos\alpha)$

$=\cos^2\alpha\ell n\cos\alpha+\sin^2\alpha\,\ell n\sin\alpha.$

3. Calculate the following limits :

निम्न सीमाऐं ज्ञात कीजिए :

(i) $\lim_{x\to\infty}\left(\frac{x^2-2x+1}{x^2-4x+2}\right)^x$ (ii) $\lim_{x\to 0}(1+\sin x)^{\cos x}$ (iii) $\lim_{x\to 0}(\cos x+a\sin bx)^{\frac{1}{x}}$

(iv) $\lim_{x\to 0}\left(\tan\left(\frac{\pi}{4}+x\right)\right)^{1/x}$ (v) $\lim_{n\to\infty}\left(4^n+5^n\right)^{1/n}$ (vi) $\lim_{x\to 5}\frac{x^2-9x+20}{x-[x]}$ ([.] denotes G.I.F.)

(vii) $\lim_{x\to -1}\frac{\sqrt{\pi}-\sqrt{\arccos x}}{\sqrt{x+1}}$ (viii) $\lim_{x\to\infty}\left(\frac{1^{1/x}+2^{1/x}+3^{1/x}+....+n^{1/x}}{n}\right)^{nx}$

**Ans.** (i) $e^2$ (ii) 1 (iii) $e^{ab}$ (iv) $e^2$ (v) 5

(vi) L.H.L. = 0, R.H.L. = 1, D.N.E. (vii) $1/\sqrt{2\pi}$ (viii) n!

**Sol.** (i) $\lim_{x\to\infty}\left(\frac{x^2-2x+1}{x^2-4x+2}\right)^x=\exp\left(\lim_{x\to\infty}\frac{2x^2-x}{x^2-4x+2}\right)$

$=e^2$

(ii) $\lim_{x\to 0}(1+\sin x)^{\cos x}=1$

(iii) $\lim\limits_{x\to 0}\left(\cos x + a\sin bx\right)^{\frac{1}{x}}$

$$= \exp\left(\lim_{x\to 0}\left(\frac{-2\sin^2\frac{x}{2} + a\sin bx}{x}\right)\right)$$

$= \exp(ab) = e^{ab}$

(iv) $\lim\limits_{x\to 0}\left(\tan\left(\frac{\pi}{4}+x\right)\right)^{\frac{1}{x}}$

$$= \exp\left(\lim_{x\to 0}\frac{\tan\left(1+\tan\left(\frac{\pi}{4}+x\right)\right)}{x}\right) = e^2$$

(v) $\lim\limits_{n\to\infty}\left(4^n+5^n\right)^{\frac{1}{n}} = 5\lim\limits_{n\to\infty}\left(\left(\frac{4}{5}\right)^n+1\right)^{\frac{1}{n}} = 5$

(vi) $\lim\limits_{x\to 5}\frac{x^2-9x+20}{x-[x]} = \lim\limits_{x\to 5}\frac{(x-5)(x-4)}{x-[x]}$

$$\left.\begin{aligned} \text{LHL} &= \lim_{h\to 0}\frac{-h(1-h)}{(5-h-4)} = 0 \\ \text{RHL} &= \lim_{h\to 0}\frac{h(1+h)}{h} = 1 \end{aligned}\right\} \text{DNE}$$

(vii) $\lim\limits_{x\to -1}\frac{\sqrt{\pi}-\sqrt{\cos^{-1}(x)}}{\sqrt{x+1}}$

$$= \lim_{\theta\to\pi}\frac{\sqrt{\pi}-\sqrt{\theta}}{\sqrt{1+\cos\theta}} = \lim_{\theta\to\pi}\frac{\pi-\theta}{\left(\sqrt{\pi}+\sqrt{\theta}\right)\left|\sqrt{2}\cos^2\frac{\theta}{2}\right|}$$

$$\text{RHL} = \lim_{h\to 0}\frac{-h}{2\sqrt{\pi}\left|\sqrt{2}\cos\left(\frac{\pi}{2}+\frac{h}{2}\right)\right|}$$

$$= \lim_{h\to 0} \frac{-h}{2\sqrt{\pi}\cdot\sqrt{2}\sin\frac{h}{2}} = \frac{1}{2\sqrt{2}\pi}$$

$$\text{LHL } \lim_{h\to 0} \frac{h}{2\sqrt{\pi}\left|\sqrt{2}\cos\left(\frac{\pi}{2}-\frac{h}{2}\right)\right|} = \frac{1}{2\sqrt{2}\pi}$$

$$\text{(viii) } \lim_{x\to\infty}\left(\frac{1^{\frac{1}{x}}+2^{\frac{1}{x}}+3^{\frac{1}{x}}+....+n^{\frac{1}{x}}}{n}\right)^{nx} \left(1^{\infty}\right)$$

$$= \exp\left(\lim_{x\to\infty}\left(\frac{1^{\frac{1}{x}}+.....+n^{\frac{1}{x}}-n}{n}\right)\cdot nx\right)$$

$$= \exp\left(\lim_{x\to 0}\frac{\left(1^x+....+n^x-n\right)}{x}\right)$$

$$= \exp\left(\lim_{x\to 0}\left(\left(\frac{1^x-1}{x}\right)+\left(\frac{2^x-1}{x}\right)+....+\left(\frac{n^x-1}{x}\right)\right)\right)$$

$$= \exp\left(\ln 1+\ln 2+....+\ln n\right) = \underline{\lfloor n}$$

4. If $\lim_{x\to 0}\frac{4^{3^x}-4^{2^x}}{x} = (\ln N)\left(\ln\frac{3}{2}\right)$, then N is equal to :

यदि $\lim_{x\to 0}\frac{4^{3^x}-4^{2^x}}{x} = (\ln N)\left(\ln\frac{3}{2}\right)$, तो N का मान होगा :

**Ans. 256**

**Sol.** $\lim_{x\to 0}\frac{4^{2^x}\left(4^{3^x-2^x}-1\right)}{x}$

$$= \lim_{x\to 0}\frac{4^{2^x}\left(4^{3^x-2^x}-1\right)}{3^x-2^x}\cdot\left(\frac{3^x-2^x}{x}\right)$$

$$\lim_{x\to 0}\frac{4^{2^x}\left(4^{3^x-2^x}-1\right)}{3^x-2^x}\cdot\frac{2^x\left(\left(\frac{3}{2}\right)^x-1\right)}{x}=4.\ell n4.\ell n\frac{3}{2}$$

5. (i) $\lim_{x\to 1^-}2^{-2^{(1-x)^{-1}}}$ (ii) $\lim_{x\to 1^+}2^{-2^{(1-x)^{-1}}}$ (iii) $\lim_{x\to 0}\frac{e^{1/x}-e^{-1/x}}{e^{1/x}+e^{-1/x}}$

**Ans.** (i) 0 (ii) 1 (iii) D.N.E.

**Sol.** (i) $\lim_{x\to 1^-}2^{-2(1-x)^{-1}}=\lim_{h\to 0}\frac{1}{2^{2^{\frac{1}{h}}}}=0$

(ii) $\lim_{x\to 1^+}\frac{1}{2^{2^{\frac{1}{(1-x)}}}}=\lim_{h\to 0}\frac{1}{2^{2^{-\frac{1}{h}}}}=1$

(iii) $\lim_{x\to 0}\frac{e^{\frac{2}{x}}-1}{e^{\frac{2}{x}}+1}$

$$\left.\begin{array}{rcl}\text{LHL} & = & \dfrac{e^{-\frac{2}{h}}-1}{e^{-\frac{2}{h}}+1}=-1\\ \text{RHL} & = & \dfrac{e^{\frac{2}{h}}-1}{e^{\frac{2}{h}}+1}=1\end{array}\right\}\text{DNE}$$

6. If f(x) is odd linear polynomial with f(1) = 1 then find $\lim_{x\to 0}\frac{2^{f(\tan x)}-2^{f(\sin x)}}{x^2f(\sin x)}$.

यदि f(x) एक विषम रेखीय बहुपद है तथा f(1) = 1 तो $\lim_{x\to 0}\frac{2^{f(\tan x)}-2^{f(\sin x)}}{x^2f(\sin x)}$ का मान ज्ञात कीजिए।

**Ans.** $\frac{1}{2}\ln 2$

**Sol.** $\lim_{x\to 0}\frac{2^{f(\tan x)}-2^{f(\sin x)}}{x^2f(\sin x)}$

$$\begin{cases}\because f(x) & =a+b\\ f(-x) & =-ax+b=-(ax+b)\end{cases}$$

$\Rightarrow b=0$ & $f(1)=1$

$$\left.\begin{array}{r} \Rightarrow a = 1 \\ \Rightarrow f(x) = x \end{array}\right\}$$

$$= \lim_{x\to 0} \frac{2^{\tan x} - 2^{\sin x}}{x^2 \sin x} = \lim_{x\to 0} \frac{2^{\tan x} - 2^{\sin x}}{x^3}$$

$$= \lim_{x\to 0} \frac{2^{\sin x}}{x^3}\left(2^{\tan x - \sin x} - 1\right) = \lim_{x\to 0} \frac{2^{\sin x} \ln 2 (\tan x - \sin x)}{x^3}$$

$$= \ln 2 \lim_{x\to 0} \frac{(\tan x - \sin x)}{x^3} = \ln 2 \lim_{x\to 0} \frac{\tan x - \sin x}{x^3}$$

$$= \frac{1}{2} \ln 2$$

7. If $L = \lim_{x\to 4} \left( \frac{\log_5\left(\frac{5x}{4}\right)}{\log_5\left(\frac{20}{x}\right)} \right)^{\left(\log_5\left(\frac{x}{4}\right)\right)^{-1}}$, then [L] is (where [.] greatest integer function)

यदि $L = \lim_{x\to 4} \left( \frac{\log_5\left(\frac{5x}{4}\right)}{\log_5\left(\frac{20}{x}\right)} \right)^{\left(\log_5\left(\frac{x}{4}\right)\right)^{-1}}$ ,तो [L] का मान होगा (जहाँ [.] मह ाम पूर्णांक फलन दर्शाता है)

**Ans.** 7

**Sol.** $L = \lim_{x\to 4} \left[ \frac{\log_5\left(\frac{5x}{4}\right)}{\log_5\left(\frac{20}{x}\right)} \right]^{\frac{1}{\log_5\left(\frac{x}{4}\right)}} \quad \left(1^{\infty} \text{ form}\right)$

$$= \exp\left( \lim_{x\to 4} \frac{\log_5\left(\frac{5x\cdot x}{4\cdot 20}\right)}{\log_5\left(\frac{20}{x}\right)\cdot \log_5\left(\frac{x}{4}\right)} \right)$$

$$= \exp\left( \frac{2}{\log_5 5} \right)$$

$= e^2$

$\Rightarrow [L] = 7$

8. $\lim\limits_{x\to 1}\left(\tan\frac{\pi x}{4}\right)^{\tan\frac{\pi x}{2}}$

**Ans.** $e^{-1}$

**Sol.** $\lim\limits_{x\to 1}\left(\tan\frac{\pi x}{4}\right)^{\tan\frac{\pi x}{2}}$

$$= \exp\left(\lim_{x\to 1}\left(\tan\frac{\pi}{4}x-1\right)\left(\tan\frac{\pi x}{2}\right)\right)$$

$$= \exp\left(\lim_{x\to 1}\tan\left(\frac{\pi x}{4}-\frac{1}{\frac{\pi}{4}}\right)\left(1+\tan\left(\frac{\pi x}{4}\right)\right)\left(\tan\frac{\pi x}{2}\right)\right)$$

$$= \exp\left(2\lim_{x\to 1}\frac{\left(\frac{\pi x}{4}-\frac{\pi}{4}\right)}{\tan\frac{\pi x}{2}}\right)$$

$$= \exp\left(2\lim_{x\to 0}\frac{\frac{\pi x}{4}}{\cot\left(\frac{\pi x}{2}-\frac{\pi}{2}\right)}\right)$$

$$= \exp\left(2\lim_{x\to 0}\frac{\frac{\pi x}{4}}{-\tan\left(\frac{\pi x}{2}\right)}\right)$$

$$= \exp\left(-\lim_{x\to 0}\frac{\frac{\pi x}{2}\cdot\cos\left(\frac{\pi x}{2}\right)}{\sin\left(\frac{\pi x}{2}\right)}\right) = e^{-1}$$

9. The value of $\lim\limits_{x\to 0}\frac{\sin\left(\ell n\,(1+x)\right)}{\ell n\,(1+\sin x)}$ is :

$\lim\limits_{x\to 0}\frac{\sin\left(\ell n\,(1+x)\right)}{\ell n\,(1+\sin x)}$ का मान है :

(A) 0 (B) 1/2 (C) 1/4 (D) 1

**Ans. D**

**Sol.** $\lim\limits_{x\to 0} \frac{\sin\left(\ell n\,(1+x)\right)}{\ell n\,(1+\sin x)} = \lim\limits_{x\to 0} \frac{\sin\left(\ell n\,(1+x)\right)}{\ell n\,(1+x)} \cdot \frac{\sin x}{\ell n\,(1+\sin x)} \cdot \frac{\ell n(1+x)}{x} \cdot \frac{x}{\sin x} = 1 \,.1\,.1\,.1 = 1$

10. $\underset{x\to 0}{\text{Limit}} \frac{\sin\left(6x^2\right)}{\ell n \cos\left(2x^2 - x\right)} =$

(A) 12 (B) – 12 (C) 6 (D) – 6

**Ans. B**

**Sol.** $\lim\limits_{x\to 0} \frac{\sin(6x^2)}{\ell n \cos(2x^2 - x)} = \lim\limits_{x\to 0} \frac{\sin 6x^2}{6x^2} \cdot \frac{6x^2}{\ell n(\cos(2x^2 - x))}$

$= 1.\ \lim\limits_{x\to 0} \frac{6x^2}{\ell n(\cos(2x^2 - x))} \qquad \left(\frac{0}{0}\text{ form}\right) \qquad \left(\frac{0}{0}\text{ form}\right)$

$= \lim\limits_{x\to 0} \frac{12x}{\frac{(-\sin(2x^2 - x))(4x-1)}{\cos(2x^2 - x)}} = \lim\limits_{x\to 0} \frac{-12\cos(2x^2 - x)}{4x-1} \cdot \lim\limits_{x\to 0} \frac{x}{\sin(2x^2 - x)}$

$= \left(\frac{-12}{-1}\right) \cdot \lim\limits_{x\to 0} \frac{1}{(\cos(2x^2 - x))(4x-1)} = 12\ \cdot \left(\frac{1}{-1}\right) = -12$

11. If $L = \lim\limits_{x\to 0} \frac{\ln(1+3\sin^2 x)(e^{2x} - 2e^x + 1)\sin(x+1)}{(1-\cos 5x)(\sin^{-1}x)^2(x+1)}$, then $\frac{25L}{6}$ is :

(A) Less than sin 2 (B) Greater than sin 2

(C) Equal to sin 2 (D) is equal to 1

यदि $L = \lim\limits_{x\to 0} \frac{\ln(1+3\sin^2 x)(e^{2x} - 2e^x + 1)\sin(x+1)}{(1-\cos 5x)(\sin^{-1}x)^2(x+1)}$, तो $\frac{25L}{6}$ होगा :

(A) sin 2 से कम (B) sin 2 से अधिक

(C) sin 2 के बराबर (D) 1 के बराबर

**Ans. A**

**Sol.** $L = \lim\limits_{x\to 0} \frac{\ln\left(1+3\sin^2 x\right)\left(e^{2x} - 2e^x + 1\right)\sin\left(x+1\right)}{\left(1-\cos 5x\right)\left(\sin^{-1}x\right)^2\left(x+1\right)}$

$$\Rightarrow L=\lim_{x\to 0}\frac{3\sin^2 x}{2\sin^2\left(\frac{5x}{2}\right)}\cdot\frac{x^2}{\left(\sin^{-1}x\right)^2}\cdot\frac{\sin 1}{1}\cdot\frac{\left(\frac{5x}{2}\right)^2}{\left(\frac{5x}{2}\right)^2}$$

$$\Rightarrow L=\lim_{x\to 0}\frac{3\sin^2 x}{2\cdot\frac{25}{4}\cdot x^2}\cdot\sin 1=\frac{13}{50}\cdot\sin 1$$

$$\Rightarrow \frac{25L}{6}=\frac{25}{6}\cdot\frac{12}{50}\cdot\sin 1=\sin 1$$

which is less than $\sin 2$.

12. If $\lim\limits_{x\to\lambda}\left(2-\frac{\lambda}{x}\right)^{\lambda\tan\left(\frac{\pi x}{2\lambda}\right)}=\frac{1}{e}$, then $\lambda$ is equal to :

यदि $\lim\limits_{x\to\lambda}\left(2-\frac{\lambda}{x}\right)^{\lambda\tan\left(\frac{\pi x}{2\lambda}\right)}=\frac{1}{e}$, तो $\lambda$ होगा :

(A) $-\pi$ (B) $\pi$ (C) $\pi/2$ (D) $-\frac{2}{\pi}$

**Ans.** **C**

**Sol.** $\lim\limits_{x\to\lambda}\left(2-\frac{\lambda}{x}\right)^{\lambda\tan\left(\frac{\pi x}{2\lambda}\right)}$ ($1^\infty$ form)

$$=\exp\left(\lim_{x\to\lambda}\left(1-\frac{\lambda}{x}\right)\cdot\lambda\tan\left(\frac{\pi x}{2\lambda}\right)\right)$$

$$=\exp\left(\lim_{x\to\lambda}\frac{\lambda}{x}(x-\lambda)\cdot\tan\left(\frac{\pi x}{2\lambda}\right)\right)$$

$$=\exp\left(\lim_{x\to 0}\frac{\lambda x}{x+\lambda}\cdot\tan\left(\frac{\pi}{2\lambda}(x-\lambda)\right)\right)$$

$$=\exp\left(-\lambda\lim_{x\to 0}x\cdot\cot\left(\frac{\pi x}{2\lambda}\right)\right)=\exp\left(-\frac{2\lambda}{\pi}\right)$$

$$=\exp\left(-\frac{2\lambda}{\pi}\right)$$

$\Rightarrow \lambda = \frac{\pi}{2}$

13. If $\alpha_n \in \left(0, \frac{\pi}{4}\right)$ is a root of the equation $\tan\alpha + \cot\alpha = n$, $n \geq 2$, then $\lim\limits_{n\to\infty}\left(\sin\alpha_n + \cos\alpha_n\right)^n$ :

यदि $\alpha_n \in \left(0, \frac{\pi}{4}\right)$ समीकरण $\tan\alpha + \cot\alpha = n$, $n \geq 2$ का एक मूल है तो $\lim\limits_{n\to\infty}\left(\sin\alpha_n + \cos\alpha_n\right)^n$ का मान होगा :

(A) 0 (B) 1 (C) 2 (D) e

**Ans. D**

**Sol.** $\tan x + \cot x = n$

$\Rightarrow \frac{1}{\cos x \sin x} = n$

Now, $L = \lim\limits_{n\to\infty}\left(\sin\theta + \cos\theta\right)^n = \lim\limits_{n\to\infty}\left[\left(\sin\theta + \cos\theta\right)^2\right]^{\frac{n}{2}} = \lim\limits_{n\to\infty}\left[1 + 2\sin\theta\cos\theta\right]^{\frac{n}{2}}$

$= \lim\limits_{n\to\infty}\left[1 + \frac{2}{n}\right]^{\frac{n}{2}}$

it is of the form $1^\infty$

Therefore, $L = e^{\lim_{n\to\infty}\left[1+\frac{2}{n}-1\right]^{\frac{n}{2}}} = e$

1. $\lim\limits_{x\to -1}\dfrac{\cos 2-\cos 2x}{x^2-|x|}$

**Ans.** $2\sin 2$

**Sol.** $\lim\limits_{x\to -1}\dfrac{\cos 2-\cos 2x}{x^2-|x|}=\lim\limits_{x\to -1}\dfrac{2\sin\left(\dfrac{2+2x}{2}\right)\sin\left(\dfrac{2x-2}{2}\right)}{x^2+x}$

$=\lim\limits_{x\to -1}\dfrac{2\sin(x+1)\sin(x-1)}{(x+1)x}$

$=2\sin 2$

2. If $\alpha=\lim\limits_{x\to 0}\dfrac{\tan\sqrt[3]{x}.\ln(1+3x)}{\left(\tan^{-1}\sqrt{x}\right)^2\left(e^{\sqrt[3]{x}}-1\right)}$ and $\beta=\lim\limits_{x\to 0^-}\left(\ln\left(\{x\}+|[x]|\right)\right)^{\{x\}}$ then $e^{\alpha\beta}$ is. Where [.] denotes greatest integer function and {x} denotes fractional part function.

यदि $\alpha=\lim\limits_{x\to 0}\dfrac{\tan\sqrt[3]{x}.\ln(1+3x)}{\left(\tan^{-1}\sqrt{x}\right)^2\left(e^{\sqrt[3]{x}}-1\right)}$ तथा $\beta=\lim\limits_{x\to 0^-}\left(\ln\left(\{x\}+|[x]|\right)\right)^{\{x\}}$ तो $e^{\alpha\beta}$ का मान होगा। (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है , {.} भिन्नात्मक भाग दर्शाता है)

**Ans.** 8

**Sol.** $\alpha=\lim\limits_{x\to 0}\left(\dfrac{\tan x^{1/3}}{x^{1/3}}\right)\left(\dfrac{\ell n(1+3x)}{3x}\right)\left(\dfrac{3x}{\left(\tan^{-1}\sqrt{x}\right)^2}\left(\dfrac{x^{1/3}}{e^{x1/3}-1}\right)\right)=3$

$\beta=\lim\limits_{x\to 0^-}\left(\ell n\{x\}+|[x]|\right)^{\{x\}}=\left(\ell n(1+1)\right)^1=\ell n2$

$e^{3\ell n2}=8$

3. $\lim\limits_{y\to 0}\dfrac{(x+y)\sec(x+y)-x\sec x}{y}$

**Ans.** $\sec x(1+x\tan x)$

**Sol.** $\lim\limits_{y\to 0}\dfrac{(x+y)\sec(x+y)-x\sec x}{y}$

$$= \lim_{y\to 0}\frac{(x+y)\cos x - x\cos(x+y)}{\cos(x+y)\cos xy}$$

$$= \frac{1}{\cos^2 x}\lim_{y\to 0}\frac{x\cos x + y\cos x - x\cos x\cos y + x\sin x\sin y}{y}$$

$$= \frac{1}{\cos^2 x}\lim_{y\to 0}\left[\frac{x\cos x\cdot 2\sin^2\frac{y}{2}}{y} + \frac{y\cos x}{y} + \frac{x\sin x\sin y}{y}\right]$$

$$= \frac{1}{\cos^2 x}(\cos x + x\sin x)$$

$$= \sec x\,(1 + x\tan x)$$

4. If $f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$, find $\lim_{x\to 0} f(x)$

यदि $f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$ है,तो $\lim_{x\to 0} f(x)$ ज्ञात कीजिए।

**Ans.** 0

**Sol.** $f(x) = \begin{cases} x^2 & x \in Q \\ 0 & x \in Q^C \end{cases}$

$$\Rightarrow \lim_{x\to 0} f(x) = 0$$

5. If $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k^2+k}{n^3+k}$ can be expressed as rational $\frac{p}{q}$ in the lowest form, find $(p+q)$.

यदि $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k^2+k}{n^3+k}$ को $\frac{p}{q}$ रूप (सरलतम) में व्यक्त किया जा सकता है। तो $(p+q)$ का मान ज्ञात कीजिए।

**Ans.** 4

**Sol.** $$\sum_{k=1}^{n}\frac{k(k+1)}{n^3+k} = \frac{2}{n^3+1} + \frac{6}{n^3+2} + \ldots + \frac{n^2+n}{n^3+n}$$

$$\leq \lim_{n\to\infty}\sum_{k=1}^{n}\frac{k(k+1)}{n^3+k} = \leq \lim_{n\to\infty}\sum_{k=1}^{n}\frac{k^2+k}{n^3+1}$$

$$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k(k+1)}{n^3+n}$$

$$\sum_{k=1}^{n}k^2+k=\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}$$

$$\Rightarrow \lim_{n\to\infty}\sum_{k=1}^{n}\frac{k^2+k}{n^3+n}=\frac{1}{3}$$

$$\& \lim_{n\to\infty}\sum_{k=1}^{n}\frac{k^2+k}{n^3+1}=\frac{1}{3}$$

$\therefore$ By sandwitch $\frac{p}{q}=\frac{1}{3}$

$\Rightarrow p + q = 4$

6. Let $l=\lim_{x\to2}\frac{[x]}{\{x\}}$ and $m=\lim_{x\to2}\frac{\{x\}}{[x]}$ where [y] and {y} denotes largest integer less than or equal to y and fractional part of y respectively. Then which one of the following alternative is correct ?

(A) $l$ exist but m does not (B) m exist but $l$ does not

(C) Both $l$ and m exist (D) Neither $l$ nor m exist

माना $l=\lim_{x\to2}\frac{[x]}{\{x\}}$ तथा $m=\lim_{x\to2}\frac{\{x\}}{[x]}$ जहाँ [y] तथा {y} महत्तम पूर्णांक फलन तथा भिन्नात्मक भाग फलन दर्शाता है। तो निम्न में से कौनसा विकल्प सही है?

(A) $l$ विद्यमान है परन्तु m नहीं (B) m विद्यमान है परन्तु $l$ नहीं

(C) $l$ तथा m दोनों विद्यमान है (D) $l$ तथा m दोनों विद्यमान नही है

**Ans.** D

**Sol.** $\ell=\lim_{x\to2}\frac{[x]}{\{x\}}$

$$RHL=\lim_{h\to0}\frac{[2+h]}{\{2+h\}}=\frac{2}{h}=DNE$$

$$\left.\begin{aligned} \text{LHL} &= \lim_{h\to 0}\frac{1}{1-h} = 1 \\ m &= \lim_{h\to 0}\frac{\{x\}}{[x]} \\ \text{RHL} &= \lim_{h\to 0}\frac{h}{2} = 0 \end{aligned}\right\} \Rightarrow \text{Neither } \ell \text{ nor m exists}$$

$$\text{LHL} = \lim_{h\to 0}\frac{1-h}{-h} = \text{DNE}$$

7. $\lim\limits_{x\to \pi/2}\left[\dfrac{x-\frac{\pi}{2}}{\cos x}\right]$ is : (where [.] represents greatest integer function :

(A) – 1 (B) 0

(C) – 2 (D) does not exist

$\lim\limits_{x\to \pi/2}\left[\dfrac{x-\frac{\pi}{2}}{\cos x}\right]$, जहाँ [.] महत्तम पूर्णांक फलन को प्रदर्शित करता हैं, का मान है :

(A) – 1 (B) 0

(C) – 2 (D) विद्यमान नहीं है।

**Ans.** C

**Sol.** $\sinh < h < \tanh \quad , \qquad h \in \left(0, \dfrac{\pi}{2}\right)$

$$\frac{h}{\sinh} > 1 \qquad \Rightarrow \qquad \frac{-h}{\sin h} < -1$$

$$\text{LHL} = \lim_{h\to 0}\left[\frac{\frac{\pi}{2}-h-\frac{\pi}{2}}{\cos\left(\frac{\pi}{2}-h\right)}\right] = \lim_{h\to 0}\left[\frac{-h}{\sinh}\right] = -2$$

$$\text{RHL} = \lim_{h\to 0}\left[\frac{\frac{\pi}{2}+h-\frac{\pi}{2}}{\cos\left(\frac{\pi}{2}+h\right)}\right] = \lim_{h\to 0}\left[\frac{-h}{\sinh}\right] = -2 \qquad \therefore \qquad \text{LHL} = \text{RHL} = -2$$

8. $\lim\limits_{x\to\frac{\pi}{2}} \sin^{-1}\left(\dfrac{\sin(3\tan^3 x+5\tan x+1)}{(3\tan^3 x+5\tan x+1)}\right)$ is equal to :

$\lim\limits_{x\to\frac{\pi}{2}} \sin^{-1}\left(\dfrac{\sin(3\tan^3 x+5\tan x+1)}{(3\tan^3 x+5\tan x+1)}\right)$ का मान होगा :

(A) 1 (B) $\pi/4$ (C) 0 (D) $\pi/2$

**Ans.** C

**Sol.** $\lim\limits_{x\to\frac{\pi}{2}} \sin^{-1}\left(\dfrac{\sin(3\tan^3 x+5\tan x+1)}{(3\tan^3 x+5\tan x+1)}\right)$

If $x \to \dfrac{\pi^+}{2}$ then $3\tan^3 x + 5\tan x + 1 \to \infty$

$\therefore$ R.H.L = 0

If $x \to \dfrac{\pi^-}{2}$ then $3\tan^3 x + 5\tan x + 1 \to \infty$

$\Rightarrow \dfrac{\sin\left(3\tan^3 x+5\tan x+1\right)}{3\tan^3 x+5\tan x+1} \to 0$

$\therefore$ L.H.L = 0

$\therefore$ Given limit = 0.

9. $\lim\limits_{x\to 0}\left(\left[\dfrac{-5\sin x}{x}\right]+\left[\dfrac{6\sin x}{x}\right]\right)$ (where [.] denotes greatest integer function) is equal to :

$\lim\limits_{x\to 0}\left(\left[\dfrac{-5\sin x}{x}\right]+\left[\dfrac{6\sin x}{x}\right]\right)$ का मान होगा : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) 0 (B) –12 (C) 1 (D) 2

**Ans.** A

**Sol.** $L_1 = \lim\limits_{x\to 0}\left[\dfrac{-5\sin x}{x}\right] = -5$

$L_2 = \lim\limits_{x\to 0}\left[\dfrac{6\sin x}{x}\right] = 5$

$\therefore L_1 + L_2 = 0$

10. $\lim_{x\to 0}\left[\frac{\sin(\text{sgn}x)}{\text{sgn}\,x}\right]$ (where [.] denotes greatest integer function) :

(A) 0 (B) 1

(C) –1 (D) does not exist

$\lim_{x\to 0}\left[\frac{\sin(\text{sgn}x)}{\text{sgn}\,x}\right]$ का मान होगा : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) 0 (B) 1

(C) –1 (D) विद्यमान नही है

**Ans.** A

**Sol.** $\lim_{x\to 0}\left[\frac{\sin(\text{sgn}\,x)}{\text{sgn}\,x}\right]=0$ $\quad\left\{\frac{\sin(\text{sgn}\,x)}{\text{sgn}\,x}=\sin x\right.$

11. $\lim_{x\to 0}\left[\log(1+x)\frac{\tan x}{|x|}\right]$ is equal to : (where [.] represents greatest integer function)

(A) 0 (B) 1 (C) –1 (D) does not exist

$\lim_{x\to 0}\left[\log(1+x)\frac{\tan x}{|x|}\right]$ का मान होगा : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) 0 (B) 1

(C) –1 (D) कोई विद्यमान नहीं है

**Ans.** A

**Sol.** $\lim_{x\to 0}\left[\log(1+x)\frac{\tan x}{|x|}\right]$

$\lim_{x\to 0^+}\left[\log(1+x)\frac{\tan x}{x}\right]=\left[0^+.1^+\right]=0$

$\lim_{x\to 0^+}\left[\log(1+x).\frac{\tan x}{(-x)}\right]=\left[0^-.\frac{1^+}{(-)}\right]=0$

12. $\lim_{x\to 0}\left[(1-e^x)\frac{\sin x}{|x|}\right]$ is equal to : (where [.] denotes greatest integer function)

(A) 0 (B) 1

(C) –1 (D) does not exist

$\lim_{x\to 0}\left[(1-e^x)\frac{\sin x}{|x|}\right]$ का मान होगा : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है) :

(A) 0 (B) 1

(C) –1 (D) विद्यमान नहीं है

**Ans.** C

**Sol.** $\text{R.H.L.} = \lim_{x\to 0^+}\left[(1-e^x)\frac{\sin x}{x}\right]$

when $x \in (0, h)$ and $h \to 0$ then $(1 - e^x) \in (-1, 0)$ and $\frac{\sin x}{x} < 1$

So $-1 < (1 - e^x)\ \frac{\sin x}{x} < 0$ ; $\lim_{x\to 0^+}\left[(1-e^x)\frac{\sin x}{x}\right] = -1$

$\text{L.H.L.} = \lim_{x\to 0^-}\left[(1-e^x)\frac{\sin x}{-x}\right] = \lim_{x\to 0^-}\left[(e^x-1)\frac{\sin x}{x}\right]$

when $x \in (-h, 0)$ and $h \to 0$, then $e^x - 1 \in (-1, 0)$ and $\frac{\sin x}{x} < 1$

So $-1 < (e^x - 1)\ \frac{\sin x}{x} < 0$

So $\lim_{x\to 0^-}\left[(e^x-1)\frac{\sin x}{x}\right] = -1$

$\text{L.H.L.} = \text{R.H.L.} = -1$

**Hindi.** दाँयीं सीमा $= \lim_{x\to 0^+}\left[(1-e^x)\frac{\sin x}{x}\right]$

जब $x \in (0, h)$ एवं $h \to 0$, तब $(1 - e^x) \in (-1, 0)$ और $\frac{\sin x}{x} < 1$

इसलिए $-1 < (1 - e^x)\ \frac{\sin x}{x} < 0$ ; $\lim_{x\to 0^+}\left[(1-e^x)\frac{\sin x}{x}\right] = -1$

बाँयी सीमा $= \lim_{x\to 0^-}\left[(1-e^x)\frac{\sin x}{-x}\right] = \lim_{x\to 0^-}\left[(e^x-1)\frac{\sin x}{x}\right]$

जब $x \in (-h, 0)$ एवं $h \to 0$, तब $e^x - 1 \in (-1, 0)$तथा $\frac{\sin x}{x} < 1$

इसलिए $-1 < (e^x - 1)\ \frac{\sin x}{x} < 0$

इसलिए $\lim\limits_{x\to0^-}\left[(e^x-1)\dfrac{\sin x}{x}\right]=-1$

बाँयी सीमा = दाँयीं सीमा. = –1

13. $\lim\limits_{x\to1}\left[\cos ec\dfrac{\pi x}{2}\right]^{-\frac{1}{(1-x)}}$ is equal to : (where [.] denotes greatest integer function)

(A) 0 (B) 1

(C) ∞ (D) does not exist

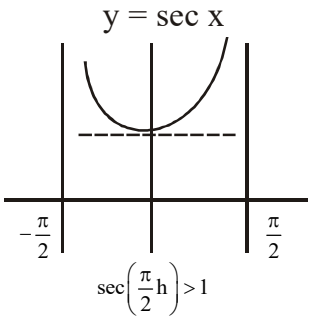
$\lim\limits_{x\to1}\left[\cos ec\dfrac{\pi x}{2}\right]^{-\frac{1}{(1-x)}}$ का मान होगा : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) 0 (B) 1

(C) ∞ (D) विद्यमान नहीं है

**Ans.** B

**Sol.** $\lim\limits_{x\to1}\left[\cos ec\dfrac{\pi x}{2}\right]^{-\frac{1}{1-x}}$




$=\lim\limits_{h\to0}\left[\cos ec\left(\dfrac{\pi}{2}(h+1)\right)\right]^{\frac{1}{h}}$

$=\lim\limits_{h\to0}\left[\sec\left(\dfrac{\pi}{2}h\right)\right]^{\frac{1}{h}}$ {1∞ form}

$RHL=\lim\limits_{h\to0}\left[\sec\left(\dfrac{\pi}{2}h\right)\right]^{+\frac{1}{h}}$

$LHL=\lim\limits_{h\to0}\left[\sec\left(\dfrac{\pi}{2}h\right)\right]^{-\frac{1}{h}}=0$

$RHL=\exp\left(\lim\limits_{h\to0}\dfrac{\left[\sec\left(\dfrac{\pi}{2}h\right)\right]-1}{h}\right)=1$

14. $\lim_{x\to 0} |x|^{[\cos x]}$ is equal to : (where [.] denotes greatest integer function)

(A) 0 (B) 1

(C) does not exist (D) e

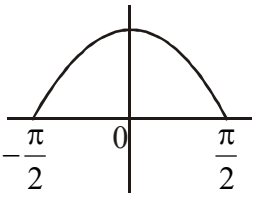
$\lim_{x\to 0} |x|^{[\cos x]}$ का मान होगा : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) 0 (B) 1

(C) विद्यमान नहीं है (D) e

**Ans.** B

**Sol.** $\lim_{h\to 0} |x|^{[\cos x]} = 1$




15. $\lim_{x\to 0}\left\{(1+x)^{\frac{2}{x}}\right\}$ : (where {.} denotes fractional part function)

(A) $e^2 - 7$ (B) $e^2 - 8$

(C) $e^2 - 6$ (D) None of these

$\lim_{x\to 0}\left\{(1+x)^{\frac{2}{x}}\right\}$ का मान होगा : (जहाँ {.} भिन्नात्मक भाग दर्शाता है)

(A) $e^2 - 7$ (B) $e^2 - 8$

(C) $e^2 - 6$ (D) इनमें से कोई नहीं

**Ans.** A

**Sol.** $\lim_{x\to 0}\left\{(1+x)^{\frac{2}{x}}\right\} = \lim_{x\to 0}\left((1+x)^{\frac{2}{x}} - \left[(1+x)^{\frac{2}{x}}\right]\right) = e^2 - 7$

16. Let $f(x) = \frac{\tan x}{x}$, then the value of $\lim_{x\to 0}\left(f([x]) + x^2\right)^{\frac{1}{\{f(x)\}}}$ is equal to : (where [.], {.} denotes greatest integer function and fractional part functions respectively)

(A) $e^{-3}$ (B) $e^3$ (C) $e^2$ (D) non-exitent

माना $f(x) = \frac{\tan x}{x}$, तो $\lim_{x\to 0}\left(f([x]) + x^2\right)^{\frac{1}{\{f(x)\}}}$ का मान होगा : (जहाँ [.] तथा {.} महत्तम पूर्णांक तथा भिन्नात्मक भाग फलन दर्शाते है)

(A) $e^{-3}$ (B) $e^3$ (C) $e^2$ (D) विद्यमान नहीं है

**Ans.** D

**Sol.** $f(x)=\dfrac{\tan x}{x}$

$\lim\limits_{x\to 0}\left(f([x])+x^2\right)\{f(x)\}$

$\text{LHL}=\lim\limits_{h\to 0}\left(f(-1)+h^2\right)\{f(-h)\}$

$=\lim\limits_{h\to 0}\left(h^2+\tan 1\right)\left(\dfrac{\tanh}{h}-0\right)$

$=\lim\limits_{h\to 0}\left(h^2+\tan 1\right)\dfrac{h}{\tanh}=\tan 1$

$\text{RHL}=\lim\limits_{h\to 0}\underbrace{\left(f(0)+h^2\right)}_{\text{undetermined form}}\dfrac{h}{\tanh}$

17. Let $f:R\to R$ be defined by $f(x)=\begin{cases}x^3-2x & \text{if } x\in Q\\ x^2-2 & \text{if } x\in R-Q\end{cases}$. Then sum of all possible values of $\alpha\in R$ for which $\lim\limits_{x\to\alpha}f(x)$ exists, is :

माना फलन $f:R\to R$ परिभाषित है कि $f(x)=\begin{cases}x^3-2x & \text{if } x\in Q\\ x^2-2 & \text{if } x\in R-Q\end{cases}$ तो $\alpha\ (\in R)$ के सभी संभव मानों का योगफल होगा जिसके लिए $\lim\limits_{x\to\alpha}f(x)$ विद्यमान है :

(A) $\sqrt{2}$ (B) $2\sqrt{2}$ (C) 1 (D) 0

**Ans.** C

**Sol.** $f(x)=\begin{cases}x^3-2x & x\in Q\\ x^2-2 & x\in Q^C\end{cases}$

$\therefore$ limit exists

if $x^3-2x=x^2-2$

$x^3-x^2-2x+2=0$

$(x^2-2)(x-1)=0$

$x=1$ or $x=\sqrt{2}$ or $x=-\sqrt{2}$

sum = 1

18. The value of $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^2}{n^3+n^2+r}$ equals :

$\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^2}{n^3+n^2+r}$ का मान होगा :

(A) 1/3 (B) 1/2 (C) 2/3 (D) 1

**Ans.** A

**Sol.** $\sum_{r=1}^{n}\frac{r^2}{n^3+n^2+n}\le\sum_{r=1}^{n}\frac{r^2}{n^3+n^2+r}\le\sum_{r=1}^{n}\frac{r^2}{n^3+n^2+1}$

$$\Rightarrow \lim_{n\to\infty}\frac{n(n+1)(2n+1)}{6\left(n^3+n^2+n\right)}\le\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^2}{n^3+n^2+r}\le\lim_{n\to\infty}\frac{n(n+1)(2n+1)}{6\left(n^3+n^2+1\right)}$$

$$\downarrow \qquad\qquad\qquad \downarrow$$

$$\frac{1}{3} \qquad\qquad\qquad \frac{1}{3}$$

19. If $A=\lim_{x\to0}\frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ and $B=\lim_{x\to0}\frac{[|x|]}{x}$, then : (where [.] denotes greatest integer function)

(A) A = 1 (B) A does not exist

(C) B = 0 (D) B = 1

यदि $A=\lim_{x\to0}\frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ तथा $B=\lim_{x\to0}\frac{[|x|]}{x}$, तो : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) A = 1 (B) A विद्यमान नहीं है

(C) B = 0 (D) B = 1

**Ans.** BC

**Sol.** $A=\lim_{x\to0}\frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$

$$\lim_{x\to0^-}\frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}=\lim_{x\to0^-}\frac{x}{-x}=-1$$

$$\lim_{x\to0^+}\frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}=\lim_{x\to0^+}\frac{x}{x}=1$$

$\Rightarrow$ A does not exist

$$B = \lim_{x\to 0}\frac{[|x|]}{x} = 0 \text{ as } [|x|] = 0 \text{ when } x \to 0.$$

20. $\lim_{x\to\infty} \sin^{-1}\sin\left(\frac{x^2+2}{x^2+1}\right)$ is equal to : (where [.] and {.} denote greatest integer and fractional part function )

$\lim_{x\to\infty} \sin^{-1}\sin\left(\frac{x^2+2}{x^2+1}\right)$ का मान होगा : (जहाँ [.] तथा {.} महत्तम पूर्णांक तथा भिन्नात्मक भाग फलन दर्शाता है)

(A) $\lim_{x\to\infty}\left[\frac{x^2+2}{x^2+1}\right]$

(B) $\lim_{x\to\infty}\frac{x^2+2}{x^2+1}$

(C) $\lim_{x\to\infty}\left\{\frac{x^2+2}{x^2+1}\right\}$

(D) $\lim_{x\to\infty}\left\{\frac{x^2}{x^2+1}\right\}$

**Ans.** ABD

**Sol.** $\lim_{x\to\infty} \sin^{-1}\left(\sin\left(\frac{x^2+2}{x^2+1}\right)\right)$

$$= \lim_{x\to\infty} \sin^{-1}\left(\sin\underbrace{\left(1+\frac{1}{x^2+1}\right)}_{\in(-1,2]}\right) = 1$$

(A) $\lim_{x\to\infty}\left[\frac{x^2+2}{x^2+1}\right] = \lim_{x\to\infty}\left[1+\frac{1}{x^2+1}\right] = 1$

(B) $\lim_{x\to\infty}\frac{x^2+2}{x^2+1} = 1$

(C) $\lim_{x\to\infty}\left\{\frac{x^2+2}{x^2+1}\right\} = 0$

(D) $\lim_{x\to\infty}\left\{\frac{x^2}{x^2+1}\right\} = \lim_{x\to\infty}\left(\frac{x^2}{x^2+1}-0\right) = 1$

21. Which of the following limits vanish?

निम्न में से कौनसी सीमा का मान शून्य है?

(A) $\lim_{x\to 0}\frac{x}{\cos^{-1}x}$

(B) $\lim_{x\to\infty}\frac{x\sin x!}{1+x^2}$

(C) $\lim_{x\to 0}\left(\frac{1}{\sin x}-\frac{1}{\tan x}\right)$

(D) $\lim_{n\to\infty}\frac{2^n}{n!}$

**Ans.** ABCD

**Sol.** (A) $\lim_{x\to 0}\frac{x}{\cos^{-1}x}=\frac{0}{\frac{\pi}{2}}=0$

(B) $\lim_{x\to\infty}\frac{x}{1+x^2}\underbrace{\sin\lfloor x}_{\in[-1,1]}=0$

(C) $\lim_{x\to 0}\left(\frac{1-\cos x}{\sin x}\right)=\lim_{x\to 0}\frac{2\sin^2\frac{x}{2}}{\frac{x^2}{4}}-\frac{x^2}{4\sin x}$

(D) $\lim_{n\to\infty}\frac{2^n}{\lfloor n}=\lim_{n\to\infty}\frac{\sum_{r=1}^{n}{}^nC_r}{\lfloor n}=0$

$\left\{\because 2^n={}^nC_0+{}^nC_1+.....+{}^nC_n\right\}$ $\left\{\begin{matrix}\because \lfloor n>2^n\\ \text{if } n>3\end{matrix}\right\}$

1. Let $f(x)=\begin{Bmatrix} x; x\le 0 \\ -x; x>0 \end{Bmatrix}$ $g(x)=f(x)+|x|$, Find $\lim\limits_{x\to 0^+}\left(\log_{|\sin x|} x\right)^{g(x)}$.

माना $f(x)=\begin{Bmatrix} x; x\le 0 \\ -x; x>0 \end{Bmatrix}$ $g(x)=f(x)+|x|$, $\lim\limits_{x\to 0^+}\left(\log_{|\sin x|} x\right)^{g(x)}$ का मान ज्ञात कीजिए।

**Ans.** 1

**Sol.** $g(x)=\begin{cases} 0 & x>0 \\ 0 & x\le 0 \end{cases}$

$\lim\limits_{x\to 0^+}\left(\log_{|\sin x|} x\right)^0=1$

2. Let $f(x)=\begin{Bmatrix} \sin x; & \text{where } x=\text{integer} \\ 0; & \text{otherwise} \end{Bmatrix}: g(x)=\begin{Bmatrix} x^2+1 & ; & x\ne 0,2 \\ 4 & ; & x=0 \\ 5 & ; & x=2 \end{Bmatrix}$,

find (i) $\lim\limits_{x\to 0} g(f(x))$ (ii) $\lim\limits_{x\to 0} f(g(x))$ (iii) $\lim\limits_{x\to 1} f(g(x))$ (iv) $\lim\limits_{x\to 1} g(f(x))$

माना $f(x)=\begin{Bmatrix} \sin x; & \text{जहाँ } x=\text{पूर्णांक} \\ 0; & \text{अन्यथा} \end{Bmatrix}: g(x)=\begin{Bmatrix} x^2+1 & ; & x\ne 0,2 \\ 4 & ; & x=0 \\ 5 & ; & x=2 \end{Bmatrix}$,

ज्ञात कीजिए (i) $\lim\limits_{x\to 0} g(f(x))$ (ii) $\lim\limits_{x\to 0} f(g(x))$ (iii) $\lim\limits_{x\to 1} f(g(x))$ (iv) $\lim\limits_{x\to 1} g(f(x))$

**Ans.** (i) 4 (ii) 0 (iii) 0 (iv) 4

**Sol.** $f(x)=\begin{cases} \sin x & x\in I \\ 0 & x\notin I \end{cases}$

(i) $\lim\limits_{x\to 0} g(f(x))=g(0)=4$

(ii) $\lim\limits_{x\to 0} f(g(x))=\lim\limits_{h\to 0} f(h^2+1)=0$

(iii) $\lim\limits_{x\to 1} f(g(x))=\lim\limits_{h\to 0} f((1-h)^2+1)=0$

(iv) $\lim_{x\to 1} g\left(f\left(x\right)\right) = g\left(0\right) = 4$

3. If $\lim_{x\to 0} (1 + ax + bx^2)^{2/x} = e^3$, then the find a and b.

यदि $\lim_{x\to 0} (1 + ax + bx^2)^{2/x} = e^3$, तो a तथा b ज्ञात करें।

**Ans.** $a = 3/2, b \in R$

**Sol.** $\lim_{x\to 0}\left(1+ax+bx^2\right)^{\frac{2}{x}} = \exp\left(\lim_{x\to 0}\left(a+bx\right)2\right)$

$= e^{2a} = e^3$

$\Rightarrow a = \frac{3}{2}$ & $b \in R$

4. Evaluate : $\lim_{x\to[a]} \frac{e^{\{x\}} - \{x\} - 1}{\{x\}^2}$ where $\{x\}$ denotes the fractional part of x and [a] denotes the integral part of a.

$\lim_{x\to[a]} \frac{e^{\{x\}} - \{x\} - 1}{\{x\}^2}$ का मान ज्ञात करें। (जहाँ $\{.\}$ भिन्नात्मक भाग दर्शाता है [.] महत्तम पूर्णांक फलन दर्शाता है).

**Ans.** R.H.L. = 1/2, L.H.L. = e – 2, D.N.E.

**Sol.** $\lim_{x\to[a]} \frac{e^{\{x\}} - \{x\} - 1}{\{x\}^2}$

$$LHL = \lim_{h\to 0} \frac{e^{\{[a]-h\}} - \{[a]-h\} - 1}{\{[a]-h\}^2}$$

$$= \lim_{h\to 0} \frac{e^{[a]-h-([a]-1)} - (1-h) - 1}{(1-h)^2} = e - 2$$

$$RHL = \lim_{h\to 0} \frac{e^h - h - 1}{h^2} = \frac{1}{2}$$

5. Evaluate a, b, c and d, if $\lim_{x\to\infty}\left(\sqrt{x^4+ax^3+3x^2+bx+2} - \sqrt{x^4+2x^3-cx^2+3x-d}\right) = 4$

a, b, c तथा d ज्ञात कीजिए, यदि $\lim_{x\to\infty}\left(\sqrt{x^4+ax^3+3x^2+bx+2} - \sqrt{x^4+2x^3-cx^2+3x-d}\right) = 4$

**Ans.** $a = 2, b \in R, c = 5$ and $d \in R$

**Sol.** $\lim\limits_{x\to\infty}\left(\sqrt{x^4+ax^3+3x^2+bx+2}-\sqrt{x^4+2x^3-cx^2+3x-d}\right)=4$

$$\Rightarrow \lim_{y\to0}\left\{\frac{\sqrt{1+ay+3y^2+by^3+2y^4}-\sqrt{1+2y-cy^2+3y^3-dy^4}}{y^2}\right\}=4$$

$$\Rightarrow \lim_{y\to0}\frac{(a-2)\,y+(3+c)y^2+(b-3)y^3+(2+d)y^4}{2y^2}=4$$

$$\Rightarrow \quad a-2=0 \quad \& \quad \frac{3+c}{2}=4 \quad \Rightarrow \quad c=5$$

$$\Rightarrow \quad a=2,\ c=5,\ b\in R,\ d\in R$$

6. If $\dfrac{\sin x+ae^x+be^{-x}+c\ln(1+x)}{x^3}$ has a finite limit as $x\to0$, find it and determine the values of a, b, c.

यदि $\dfrac{\sin x+ae^x+be^{-x}+c\ln(1+x)}{x^3}$ , $x\to0$ पर सीमा परिमित है तो सीमा, a, b तथा c ज्ञात कीजिए।

**Ans.** $L=-1/3,\ a=-1/2,\ b=1/2,\ c=0$

**Sol.** $\lim\limits_{x\to0}\dfrac{\sin x+ae^x+be^{-x}+c\ln(1+x)}{x^3}=\text{finite}$

$\Rightarrow a+b=0$

$$\lim_{x\to0}\frac{\left(x-\frac{x^3}{\lfloor3}+.....\right)+a\left(e^x-e^{-x}\right)+c\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+....\right)}{x^3}$$

$$=\left\{\because e^x-e^{-x}=\left(1+x\frac{x^2}{\lfloor2}+....\right)-\left(1-x\frac{x^2}{\lfloor2}+....\right)\right\}$$

$$=2x+\frac{2x^3}{\lfloor3}+....$$

$$\Rightarrow \lim_{x\to0}\frac{(2a+1+c)x+\frac{x^3}{\lfloor3}\left(-1+2a+\frac{c}{3}\right)-\frac{x^2}{2}c+....}{x^3}$$

$\Rightarrow 2a+1+xc=0 \qquad \& \ c=0$

$\Rightarrow a=-\dfrac{1}{2} \qquad \& \ L=-\dfrac{1}{3}$

$b = \frac{1}{2}$

$c = 0$

7. (i) $\lim\limits_{x\to a}\frac{a^x - x^a}{x^x - a^a} = -1$, then find the value of a. (ii) $\lim\limits_{x\to 0}\frac{e-(1+x)^{1/x}}{\tan x}$

(i) $\lim\limits_{x\to a}\frac{a^x - x^a}{x^x - a^a} = -1$, तो a का मान ज्ञात कीजिए. (ii) $\lim\limits_{x\to 0}\frac{e-(1+x)^{1/x}}{\tan x}$

**Ans.** (i) $a = 1$ (ii) $e/2$

**Sol.** (i) $\lim\limits_{x\to a}\frac{a^x - x^a}{x^x - a^a} = -1$

(ii) $\lim\limits_{x\to 0}\frac{e-(1+x)^{\frac{1}{x}}}{\tan x}$ $\left(\frac{0}{0}\text{ form}\right)$

$= \lim\limits_{x\to 0}\frac{e-(1+x)^{\frac{1}{x}}}{x}$

$= \lim\limits_{x\to 0}\frac{e-e^{\frac{1}{x}\log(1+x)}}{x}$

$= \lim\limits_{x\to 0}\frac{e\left(-1+e^{\frac{\log(1+x)}{x}-1}\right)}{x}$

$= -e\lim\limits_{x\to 0}\left(\frac{\log(1+x)-x}{x^2}\right)$

$= -e\lim\limits_{x\to 0}\frac{\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+....-x\right)}{x^2}$

$= -e\left(-\frac{1}{2}\right) = \frac{e}{2}$

8. Find a polynomial of least degree, such that $\lim_{x\to 0}\left(1+\frac{f(x)+x^2}{x^2}\right)^{1/x}=e^2$

न्यूनतम कोटि का बहुपद f(x) ज्ञात कीजिए यदि $\lim_{x\to 0}\left(1+\frac{f(x)+x^2}{x^2}\right)^{1/x}=e^2$

**Ans.** $f(x)=2x^3-x^2$

**Sol.** $\lim_{x\to 0}\left(\frac{1+f(x)+x^2}{x^2}\right)^{\frac{1}{x}}=e^2$

$\Rightarrow \lim_{x\to 0}\frac{f(x)+x^2}{x^2}=0$ & $\lim_{x\to 0}\frac{f(x)+x^2}{x^3}=2$

$\Rightarrow \lim_{x\to 0}\frac{f(x)}{x^2}=-1$

$\Rightarrow a_0=2$

$f(x)=a_0x^3-x^2$

$\Rightarrow f(x)=2x^3-x^2$

9. For a certain value of c, $\lim_{x\to\infty}\left[(x^5+7x^4+2)^C-x\right]$ is finite and non zero and has the value of equal to *l*. Find the value of $(l-2c)$.

c के एक निश्चित मान के लिये $\lim_{x\to\infty}\left[(x^5+7x^4+2)^C-x\right]$ परिमित तथा अशून्य है तथा मान *l* है. $(l-2c)$ का मान ज्ञात कीजिए।

**Ans.** 1

**Sol.** $\lim_{x\to-\infty}[(x^5+7x^4+2)^c-x]$

$=\lim_{x\to-\infty}\left(x^{5c}\left(1+\frac{7}{x}+\frac{2}{x^5}\right)^c-x\right)$

for limit to exist c = 1/5

Now to find limit put c = 1/5

$\Rightarrow \lim_{x\to-\infty}\left(x\left(1+\frac{7}{x}+\frac{2}{x^5}\right)^{1/5}-x\right)=\frac{7}{5}$ (by expansion)

10. Let $\lim_{x\to\infty} x \ln\left(e\left(1+\frac{1}{x}\right)^{1-x}\right)$ equal $\frac{m}{n}$ where m and n are relatively prime positive integer. Find (m + n).

माना $\lim_{x\to\infty} x \ln\left(e\left(1+\frac{1}{x}\right)^{1-x}\right) = \frac{m}{n}$ जहाँ m तथा n परस्पर अभाज्य है। तो (m + n) का मान ज्ञात कीजिए।

**Ans.** 5

**Sol.** $\lim_{x\to\infty} x \ln\left(e\left(1+\frac{1}{x}\right)^{1-x}\right)$

$$= \lim_{x\to\infty} \frac{1+(1-x)\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}}$$

$$= \lim_{t\to 0} \frac{1+\left(1-\frac{1}{t}\right)\ln(1+t)}{t}$$

$$= \lim_{t\to 0} \frac{1+\left(1-\frac{1}{t}\right)\left(t-\frac{t^2}{2}+\frac{t^3}{3}-\frac{t^4}{4}+....\right)}{t}$$

$$= \lim_{t\to 0} \frac{1+t-\frac{t^2}{2}+\frac{t^3}{3}-\frac{t^4}{4}+....-1+\frac{t}{2}-\frac{t^2}{3}+....}{t}$$

$= 3 + 2 \qquad \Rightarrow m + n = 5$

11. $\lim_{x\to 0}\left[\frac{2}{x^3}(\tan x - \sin x)\right]^{2/x^2}$ is :

$\lim_{x\to 0}\left[\frac{2}{x^3}(\tan x - \sin x)\right]^{2/x^2}$ का मान होगा :

(A) e (B) $\sqrt{e}$ (C) $\frac{1}{e}$ (D) $\frac{1}{\sqrt{e}}$

**Ans.** B

**Sol.** $\lim\limits_{x\to0}\left[\frac{2}{x^3}(\tan x-\sin x)\right]^{2/x^2}\Rightarrow1^\infty\Rightarrow e^{\lim\limits_{x\to0}\frac{2}{x^2}\left(\frac{2(\tan x-\sin x)}{x^3}-1\right)}$

$\Rightarrow e^{\lim\limits_{x\to0}2\left(\frac{2(\tan x-\sin x)-x^3}{x^5}\right)}\Rightarrow e^{2\lim\limits_{x\to0}\frac{2\left\{\left(x+\frac{x^3}{3}+\frac{2x^5}{15}\right)-\left\{x-\frac{x^3}{3!}+\frac{x^5}{5!}\right\}\right\}-x^3}{x^5}}\Rightarrow e^{2\lim\limits_{x\to0}\frac{2\left(\frac{2}{15}-\frac{1}{120}\right)x^5}{x^5}}$

$=e^{1/2}=\sqrt{3}$

12. If $\lim\limits_{x\to1}\sec^{-1}\left(\frac{k}{\ln x}-\frac{k}{x-1}\right)=\frac{\pi}{3}$ then k is :

यदि $\lim\limits_{x\to1}\sec^{-1}\left(\frac{k}{\ln x}-\frac{k}{x-1}\right)=\frac{\pi}{3}$ तो k होगा :

(A) 1 (B) 4 (C) –2 (D) 2

**Ans.** B

**Sol.** $\lim\limits_{x\to1}\sec^{-1}\left(\frac{k}{\ln x}-\frac{k}{x-1}\right)=\frac{\pi}{3}$

$\Rightarrow\lim\limits_{x\to1}k\left(\frac{x-1-\ln x}{(x-1)\ln x}\right)=2$

$\Rightarrow k\lim\limits_{x\to1}\left(\frac{x-1-\ln x}{(x-1)^2}\right)=2$

$\Rightarrow\frac{2}{k}=\lim\limits_{x\to1}\left[\frac{\left(x-1-\left((x-1)-\frac{(x-1)^2}{2}+\frac{(x-1)^3}{3}.....\right)\right)}{(x-1)^2}\right]$

$\Rightarrow k=4$

13. $\lim\limits_{x\to0}\frac{x(1+a\cos x)-b\sin x}{x^3}=1$ then :

$\lim\limits_{x\to0}\frac{x(1+a\cos x)-b\sin x}{x^3}=1$ तो :

(A) a = – 5/2 (B) a = –3/2, b = –1/2 (C) a = –3/2, b = –5/2 (D) a = –5/2, b = –3/2

**Ans.** D

**Sol.** $\lim\limits_{x\to 0}\dfrac{x\left(1+a\cos x\right)-b\sin x}{x^3}=1$

$$\Rightarrow \lim_{x\to 0} x\dfrac{\left(1+a\left(1-\dfrac{x^2}{\lfloor 2}+\dfrac{x^4}{\lfloor 4}-\dfrac{x^5}{\lfloor 5}+....\right)\right)-b\left(x-\dfrac{x^3}{\lfloor 3}+\dfrac{x^5}{\lfloor 5}+....\right)}{x^3}=1$$

$1 + a - b = 0$ & $-\dfrac{a}{2}+\dfrac{b}{6}=1$

$-3a + b = 6$ & $b = a + 1$

$\Rightarrow -3a + a + 1 = 6$

$-2a = 5$

$a=-\dfrac{5}{2}$

$\Rightarrow b=-\dfrac{3}{2}$

14. $\lim\limits_{x\to 0}\dfrac{\sin x^{2012}-x^{2012}\cos x^{2012}+(x^3)^{2012}}{x^{2012}\left(e^{5.x^{2012}}-1-5.x^{2012}\right)}$ is equal to :

$\lim\limits_{x\to 0}\dfrac{\sin x^{2012}-x^{2012}\cos x^{2012}+(x^3)^{2012}}{x^{2012}\left(e^{5.x^{2012}}-1-5.x^{2012}\right)}$ का मान होगा :

(A) $\dfrac{8}{75}$ (B) $\dfrac{1}{3}$ (C) $\dfrac{4}{3}$ (D) $-\dfrac{1}{3}$

**Ans.** A

**Sol.** Let $x^{2012} = t$ then $\to 0$ as $x \to 0$

$\therefore$ Given limit $=\lim\limits_{x\to 0}\dfrac{\sin t-t\cos t+t^3}{t\left(e^{5t}-1-5t\right)}$

$$\lim_{x\to 0}\dfrac{\sin t-t\cos t+t^3}{t.25t^2\left(\dfrac{e^{5t}-1-5t}{25t^2}\right)}$$

(use standard limit $\lim_{x\to 0}\frac{e^x-1-x}{x^2}=\frac{1}{2}$)

$$=\frac{2}{25}\lim_{x\to 0}\frac{\sin t-\cos t-+t^3}{t^3}$$

$$=\frac{2}{25}\lim_{x\to 0}\frac{\left(t-\frac{t^3}{3!}+\frac{t^5}{5!}....\right)-t\left(1-\frac{t^2}{2!}+\frac{t^4}{4!}\right)+t^3}{t^3}$$

$$=\frac{2}{25}.\left(\frac{1}{2}+1-\frac{1}{6}\right)=\frac{8}{75}$$

15. $\lim_{n\to\infty}\left(\frac{n^{n+1}e}{(n+1)^n}-n\right)$ is equal to :

$\lim_{n\to\infty}\left(\frac{n^{n+1}e}{(n+1)^n}-n\right)$ का मान होगा :

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) $\frac{1}{\sqrt{e}}$

**Ans.** B

**Sol.** $$\lim_{n\to\infty}\left(\frac{n^{n+1}e}{(n+1)^n}-n\right)=\lim_{n\to\infty}\left(\frac{ne}{\left(1+\frac{1}{n}\right)^n}-n\right)$$

$$=\lim_{n\to\infty}n\left(\frac{e-\left(1+\frac{1}{n}\right)^n}{\left(1+\frac{1}{n}\right)^n}\right)=\lim_{t\to 0}\frac{\left[e-(1+t)^{\frac{1}{t}}\right]}{t(1+t)^{\frac{1}{t}}}$$

$$=\lim_{t\to 0}\frac{\left[e-e^{\frac{1}{t}\ln(1+t)}\right]}{t(1+t)^{\frac{1}{t}}}=\lim_{t\to 0}\frac{-e\left[e^{\frac{\ln(1+t)}{t}-1}-1\right]}{t(1+t)^{\frac{1}{t}}}$$

$$= \lim_{t\to 0} \frac{-e\left(\ln(1+t)-t\right)}{t^2(1+t)\frac{1}{t}} = \frac{1}{2}$$

16. $\lim_{n\to\infty}\left(\left(\frac{n}{n+1}\right)^{\alpha}+\sin\frac{1}{n}\right)^{n}$ when $\alpha \in Q$ is equal to :

$\lim_{n\to\infty}\left(\left(\frac{n}{n+1}\right)^{\alpha}+\sin\frac{1}{n}\right)^{n}$ का मान होगा : (जहाँ $\alpha \in Q$)

(A) $e^{-\alpha}$ (B) $-\alpha$ (C) $e^{1-\alpha}$ (D) $e^{1+\alpha}$

**Ans.** C

**Sol.** $\lim_{n\to\infty}\left(\left(\frac{n}{n+1}\right)^{\alpha}+\sin\frac{1}{n}\right)^{n}$ [$1^{\infty}$ form]

$$= e^{\lim_{n\to\infty} n\left(\left(\frac{n}{n+1}\right)^{\alpha}+\sin\left(\frac{1}{n}\right)-1\right)} = e^{\lim_{n\to\infty}\left(n\left(\frac{n}{n+1}\right)^{\alpha}+\frac{\sin(1/n)}{(1/n)}-n\right)}$$

$$= e^{\lim_{n\to\infty}\left(n\left(1-\frac{1}{n+1}\right)^{\alpha}+1-n\right)} = e^{\lim_{n\to\infty}\left(n\left(1-\frac{\alpha}{n+1}+\ldots\ldots\right)+1-n\right)} = e^{1-\alpha}$$

17. $\lim_{n\to\infty}\frac{e^{n}}{\left(1+\frac{1}{n}\right)^{n^2}}$ equals :

$\lim_{n\to\infty}\frac{e^{n}}{\left(1+\frac{1}{n}\right)^{n^2}}$ बराबर होगा :

(A) 1 (B) ½ (C) e (D) $\sqrt{e}$

**Ans.** D

**Sol.** $\lim_{n\to\infty}\frac{e^{n}}{\left(1+\frac{1}{n}\right)^{n^2}} = \lim_{t\to 0}\frac{e^{\frac{1}{t}}}{(1+t)^{\frac{1}{t^2}}}$

$$= \lim_{t\to 0} \frac{e^{\frac{1}{t}}}{e^{\frac{1}{t^2}} \log(t+1)}$$

$$= \lim_{t\to 0} e^{\frac{1}{t}} \left(1 - \frac{\log(1+t)}{t}\right)$$

$$= \exp\left(\lim_{t\to 0} \frac{\left(t - \log(1+t)\right)}{t^2}\right)$$

$$= \exp\left(\lim_{t\to 0} \frac{\left(t - \left(t - \frac{t^2}{2} + \frac{t^3}{3} + ....\right)\right)}{t^2}\right)$$

$$\exp\left(\lim_{t\to 0}\left(\frac{1}{2} - \frac{t}{3} + ....\right)\right)$$

$$= \sqrt{e}$$

**Built in Limits**

18. Let $f(x) = \lim_{n\to\infty} \frac{1}{\left(\frac{3}{\pi}\tan^{-1} 2x\right)^{2n} + 5}$. Then the set of values of x for which f(x) = 0 is :

माना $f(x) = \lim_{n\to\infty} \frac{1}{\left(\frac{3}{\pi}\tan^{-1} 2x\right)^{2n} + 5}$. तो x के मानों का समुच्चय होगा जिसके लिए f(x) = 0 है :

(A) $|2x| > \sqrt{3}$ (B) $|2x| < \sqrt{3}$ (C) $|2x| \geq \sqrt{3}$ (D) $|2x| \leq \sqrt{3}$

**Ans.** A

**Sol.** Given, $f(x) = \lim_{x\to\infty} \frac{1}{\left(\frac{3\pi}{}\tan^{-1} 2x\right)^{2n} + 5} = 0$

or $\left[\left(\frac{3}{\pi}\tan^{-1} 2x\right)\right]^n \to \infty$

or $\left(\frac{3}{\pi}\tan^{-1}2x\right)>1$

or $\left|\tan^{-1}2x\right|>\frac{\pi}{3}$

$\Rightarrow \tan^{-1}2x<-\frac{\pi}{3}$ or $\tan^{-1}2x>\frac{\pi}{3}$

$\Rightarrow 2x<-\sqrt{3}$ or $2x>\sqrt{3}$ i.e. $|2x|>\sqrt{3}$

**Paragraph for Question 19 to 21**

Let $f(x)=\begin{cases}x^2+1, & x<1\\ x^3, & x\geq 1\end{cases}, g(x)=2^x$ and $h(x)=\frac{\sin x}{x}$

On the basis of above information, answer the following questions.

माना $f(x)=\begin{cases}x^2+1, & x<1\\ x^3, & x\geq 1\end{cases}, g(x)=2^x$ तथा $h(x)=\frac{\sin x}{x}$

उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उतर दीजिए।

19. $|f(h(0^+))-f(h(0^-))|$ is equal to :

$|f(h(0^+))-f(h(0^-))|$ का मान होगा :

(A) 3 (B) 0 (C) 1 (D) 2

**Ans.** B

**Sol.** $f\left(h\left(0^+\right)\right)=f\left(\lim_{x\to 0^+}\left(\frac{\sin x}{x}\right)\right)=f\left(1^-\right)=2$

$f\left(h\left(0^-\right)\right)=f\left(\lim_{x\to 0^-}\left(\frac{\sin x}{x}\right)\right)=f\left(1^-\right)=2$

$\left|f\left(h\left(0^+\right)\right)-f\left(h\left(0^-\right)\right)\right|=0$

20. $|f(g(0^+))-f(g(0^-))|$ is equal to :

$|f(g(0^+))-f(g(0^-))|$ का मान होगा :

(A) 3 (B) 0 (C) 1 (D) 2

**Ans.** C

**Sol.** $f\left(g\left(0^+\right)\right)=f\left(\lim_{x\to 0^+}2^x\right)=f\left(1^+\right)-1$

$$f\left(g\left(0^-\right)\right)=f\left(\lim_{x\to 0^-}2^x\right)=f\left(1^-\right)-1$$

$$\left|f\left(g\left(0^+\right)\right)-f\left(g\left(0^-\right)\right)=1\right|$$

21. $|f(g(h(0^+))) - f(g(h(0^-)))|$ is equal to :

$|f(g(h(0^+))) - f(g(h(0^-)))|$ का मान होगा :

(A) 3 (B) 0 (C) 1 (D) 2

**Ans.** B

**Sol.** $f\left(g(h(0^+)\right)=f(g(1^-))=f(2^-)=8$

$f\left(g(h(0^-)\right)=f(g(1^-))=f(2^-)=8$

$$\left|f\left(g\left(h\left(0^+\right)\right)\right)-f\left(g\left(h\left(0^-\right)\right)\right)\right|=0$$

22. The value(s) of 'n' for which $\lim_{x\to 1}\frac{e^{x-1}-x}{(x-1)^n}$ exists is/are :

'n' का मान होगा, जिसके लिए $\lim_{x\to 1}\frac{e^{x-1}-x}{(x-1)^n}$ विद्यमान है :

(A) 1 (B) 2 (C) 3 (D) 4

**Ans.** AB

**Sol.** Put $x = 1 + z$

$$\lim_{z\to 0}\frac{e^z-1-z}{z^n}=\lim_{z\to 0}\frac{1+\frac{z}{1!}+\frac{z^2}{2!}+\frac{z^3}{3!}+.....+-1-z}{z^n}$$

$$=\lim_{z\to 0}\frac{z^2\left(\frac{1}{2!}+\frac{z}{3!}+.....\right)}{z^n}$$

$\therefore$ Possible values of 'n' among options are 1,2

23. $\lim_{x\to+\infty}\left(1+\frac{1}{x^n}\right)^x$ $(n > 0)$

(A) e at $n = 1$ (B) 1 if $n > 1$ (C) $\infty$ if $0 < n < 1$ (D) 0 if $0 < n < 1$

**Ans.** ABC

**Sol.** $\lim\limits_{x\to+\infty}\left(1+\dfrac{1}{x^n}\right)^x \quad (n>0)$

$$= \exp\left(\lim_{x\to\infty}\frac{1}{x^{n-1}}\right)$$

$$= \begin{cases} e & \text{if} \quad n=1 \\ 1 & \quad\quad n>1 \\ \infty & \text{if} \quad 0<n<1 \end{cases}$$